Architecture Opportunities: Legacy CTM → RustPBX Platform

> **Document Purpose:** Step-by-step analysis of the current CallTrackingMetrics (CTM) + Twilio call tracking workflow, identifying latency, cost, and control inefficiencies at every layer, with concrete redesign opportunities for the RustPBX replacement platform.
>
> **Last Updated:** 2026-02-22

Executive Summary

Preliminary architectural analysis indicates that we can dramatically improve system functionality, decrease system latency, and reduce overall system costs (better, faster, cheaper).

The current architecture routes every customer interaction through **6+ external vendors** and **7+ disconnected data silos**, creating compounding cost, latency, and control problems at every layer. The eight redesign opportunities below eliminate these inefficiencies by bringing ownership in-house — replacing vendor-locked middleware with direct integrations and a unified PostgreSQL data layer.

Latency improvements across the stack. The legacy CTM + Twilio architecture introduces compounding delays at every layer. Replacing it with Telnyx SIP trunking and Linode-hosted RustPBX eliminates entire network hops and vendor processing stages. The table below summarizes before-and-after latency based on vendor-published specifications:

Layer	Legacy (CTM + Twilio)	Target (Telnyx + RustPBX)	Improvement
Page load (DNI number swap)	100–300ms (external CTM API round-trip per page view)	<10ms (self-hosted Linode endpoint, same-region)	10–30× faster
Call setup (post-dial delay)	5–10s typical (Twilio double-hop: PSTN GW → transcoder → SIP trunk → CTM routing)	<1s WebRTC / 2–4s PSTN via Telnyx single-hop	60–90% reduction
Audio path (per-packet RTT)	200–400ms (3 processing stages: Twilio PSTN GW → transcoder → SIP trunk → CTM → agent)	50–100ms (single SIP hop: Telnyx private network → RustPBX → agent)	75% reduction
Agent path (US → PV, Mexico)	+30–80ms RTT added (consumer ISP, no failover, no edge PoP)	+10–30ms RTT (Telnyx AnchorSite or Linode regional proxy + cellular backup)	50–60% reduction
Transcription availability	1–15 min post-call (CTM batch ASR, locked vendor)	<300ms streaming during live call (Deepgram Nova-3, 150ms first-word)	Real-time vs. batch
Management reporting	Hours (Snowflake batch ETL from 3+ vendor pipelines)	<1s (PostgreSQL materialized views, live data)	Real-time vs. batch

Cumulative system latency (end-to-end audio round-trip through full stack): The legacy architecture's multi-vendor chain — Twilio PSTN gateway, Twilio transcoder, Twilio SIP trunk, CTM call router, and Mexico ISP path — accumulates **~300–600ms round-trip audio latency** before the caller hears the agent speak. The target architecture's single-hop path — Telnyx private network direct to RustPBX on Linode delivers **~80–150ms round-trip**, well within the industry-standard 150ms one-way threshold where latency becomes imperceptible (per Twilio and Telnyx published guidelines). For WebRTC calls that bypass PSTN entirely, round-trip latency drops to **~40–80ms** — competitive with local phone calls.

Sources: Twilio PDD documentation (5s typical North America), Twilio Voice Insights FAQ (150ms one-way acceptable), Telnyx AnchorSite documentation (20ms normal VoIP), Deepgram Nova-3 benchmarks (sub-300ms streaming ASR, 150ms first-word latency).

4.1 — Own the Click: First-Party Ad Tracking API. Replaces CTM's JavaScript tag and DNI API with a self-hosted tracking endpoint. Eliminates the 100–300ms external API penalty on every page load, breaks the dependency on third-party cookies for attribution, and creates a unified click → call → outcome database. Server-side conversion APIs (Google Ads, Meta CAPI) provide cookie-independent attribution that survives ITP, ad blockers, and incognito mode.

4.2 — Own the Form: Integrated Lead Capture. Unifies form submissions and phone calls under a single session identity. Agents receive a screen pop with the caller's form data before answering — name, inquiry, and ad source all visible in one view. Eliminates the need for separate form tools (HubSpot, Typeform) and enables single-query conversion analytics across clicks, forms, and calls.

4.3 — Own the Call Path: Click-to-Call via WebRTC. Offers an internet-only call option that bypasses PSTN entirely. Call setup drops from 5–10 seconds to under 1 second. Per-minute cost drops from ~$0.0135 to $0.00 for WebRTC calls. Opus wideband audio delivers superior quality over G.711 PSTN. Video calling becomes possible — a capability the legacy stack cannot support at all. Customers stay on the landing page during the call, preserving full session context.

4.4 — Own the Intelligence: Real-Time Transcription & AI Coaching. Streams live call audio to Deepgram Nova-3 for real-time transcription visible to agents during the call — not 1–15 minutes after it ends. An LLM coaching engine surfaces suggested responses, objection handling, and compliance alerts (HIPAA/TCPA/PCI) while the agent can still act on them. Post-call, Groq Whisper Turbo generates a final high-accuracy transcript at $0.04/hour. Auto-generated summaries and disposition codes eliminate manual post-call work.

4.5 — Own the Numbers: Direct Telnyx Integration. Cuts out both CTM and Twilio from the number management chain. Inbound per-minute cost drops from ~$0.0135 (Twilio PSTN + SIP combined) to ~$0.005 (Telnyx direct) — a **63% reduction**. Numbers are provisioned and ported under our control via Telnyx API, eliminating the 2–4 week Twilio porting timeline. Single SIP trunk with built-in redundancy replaces the current double-hop architecture.

4.6 — Own the CRM Feed: Direct Zoho Integration. Replaces CTM's auto-push with a direct RustPBX → Zoho pipeline that delivers richer data: searchable transcript text, AI-generated call summaries, sentiment scores, compliance flags, and correlated form submissions. Self-hosted recording URLs eliminate the risk of broken links if CTM is decommissioned. Full control over field mapping replaces CTM's fixed schema.

4.7 — Own the Case Pipeline: Direct Flow Legal Integration. Builds a direct RustPBX → Flow Legal feed with searchable transcripts for case discovery, AI intake summaries that replace manual agent notes, and automatic compliance flagging. Self-hosted recordings and form-call correlation provide complete intake context. Reduces manual data entry for legal staff and enables transcript-based case search that the legacy stack cannot support.

4.8 — Own the Analytics: Replace Snowflake with PostgreSQL. Once all data lives in PostgreSQL (calls, clicks, forms, CRM, cases, payments), Snowflake becomes redundant. Real-time dashboards replace batch-lag reports. Full-funnel queries — ad click → call → case → payment — run in a single SQL statement. New metrics require a SQL view, not a multi-system ETL change. Eliminates Snowflake compute billing and ETL pipeline maintenance entirely.

Combined financial impact (annual vendor spend):

Vendor / Cost Center	Annual Spend	Replaced By	Target Cost
CTM (includes Twilio PSTN)	$300,000	RustPBX + Telnyx SIP	—
Zoho CRM	$200,000	Rust CRM engine + PostgreSQL	—
Flow Legal Management	$200,000	Rust case engine + PostgreSQL	—
Snowflake	$50,000	PostgreSQL materialized views	—
SysTango Development	$250,000	In-house developers	—
Total legacy SaaS	**$1,000,000/yr**		

Estimated RustPBX platform cost:

Cost Center	Annual Estimate	Notes
Linode VPS + storage	$50,000	3 servers + 5 TB storage
Telnyx PSTN	$50,000	~500K min @ $0.01/min + 100 lines
IT infrastructure staff (×2)	$100,000	Two for redundancy, not 100% utilized
Rust + SQL developers (×2)	$100,000	Platform development and maintenance
Total RustPBX platform	**$300,000/yr**	

Net Annual SAVINGS: **~$700,000/yr (70% reduction).**

Remaining variable costs (Deepgram/Groq transcription usage) are incremental and scale with call volume.

Beyond direct cost savings, the platform gains capabilities the legacy stack cannot provide at any price: real-time transcription, real-time AI agent coaching, video calling, cookie-independent attribution, and single-database full-funnel analytics.

Table of Contents

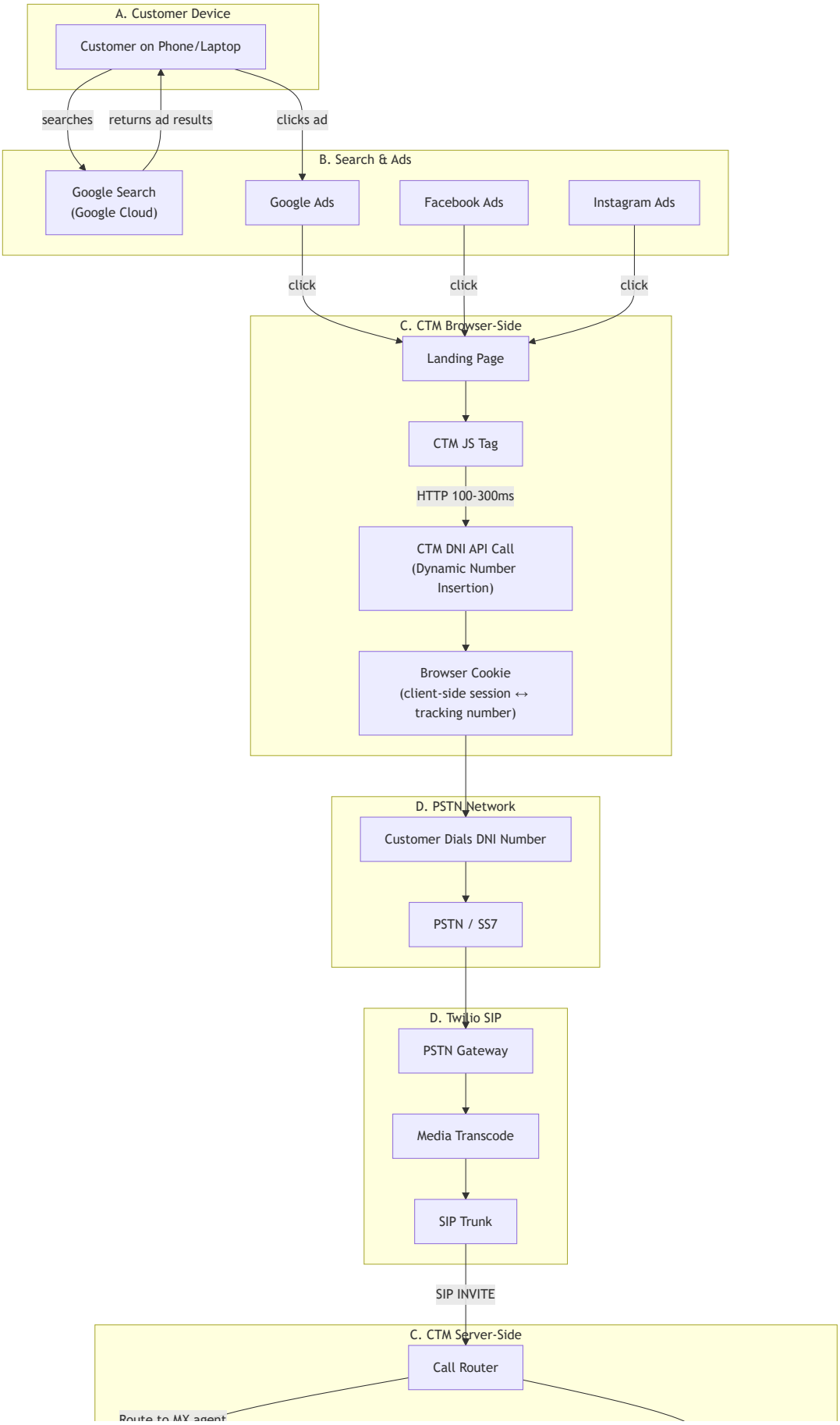
- Architecture Opportunities: Legacy CTM → RustPBX Platform
  - Executive Summary
  - Table of Contents
  - 1. Current Workflow Overview
  - 2. Step-by-Step Legacy Flow
    - Step 1: Customer Searches and Clicks an Ad (Layers A–B)
    - Step 2: Landing Page Loads CTM JavaScript (Layer C)
    - Step 3: Cookie Storage and Number Swap (Layer C)
    - Step 4: Customer Calls the Tracking Number (Layer D)
    - Step 5: Twilio Receives and Trunks the Call (Layer D)
    - Step 6: CTM Routes, Records, and Transcribes (Layer C)
    - Step 7: Agent Handles the Call (Layer E — Puerto Vallarta, Mexico)
    - Step 8: CTM Transcribes the Call (Layer F — Vendor-Locked ASR)
    - Step 9: CTM Pushes Data to Zoho CRM (Layer G)
    - Step 10: CTM Feeds Data to Flow Legal Management (Layer H)
    - Step 11: CTM and Zoho Feed Snowflake for Management Reporting (Layer I)
  - 3. Inefficiency Analysis by Layer
    - 3.1 Ad Click Attribution (Layer B)
    - 3.2 Browser-Side Tracking (Layer C)
    - 3.3 PSTN Call Path (Layer D)
    - 3.4 Twilio SIP Trunking (Layer D)
    - 3.5 CTM Platform (Layer C)
    - 3.6 Agent Experience (Layer E)
    - 3.7 Agent Geographic Location (Layer E)
    - 3.8 Transcription (Layer F)
    - 3.9 CRM Integration (Layer G — Zoho via CTM)
    - 3.10 Case Management (Layer H — Flow Legal via CTM)
    - 3.11 Management Reporting (Layer I — Snowflake)
  - 4. Redesign Opportunities
    - 4.1 Own the Click: First-Party Ad Tracking API
    - 4.2 Own the Form: Integrated Lead Capture

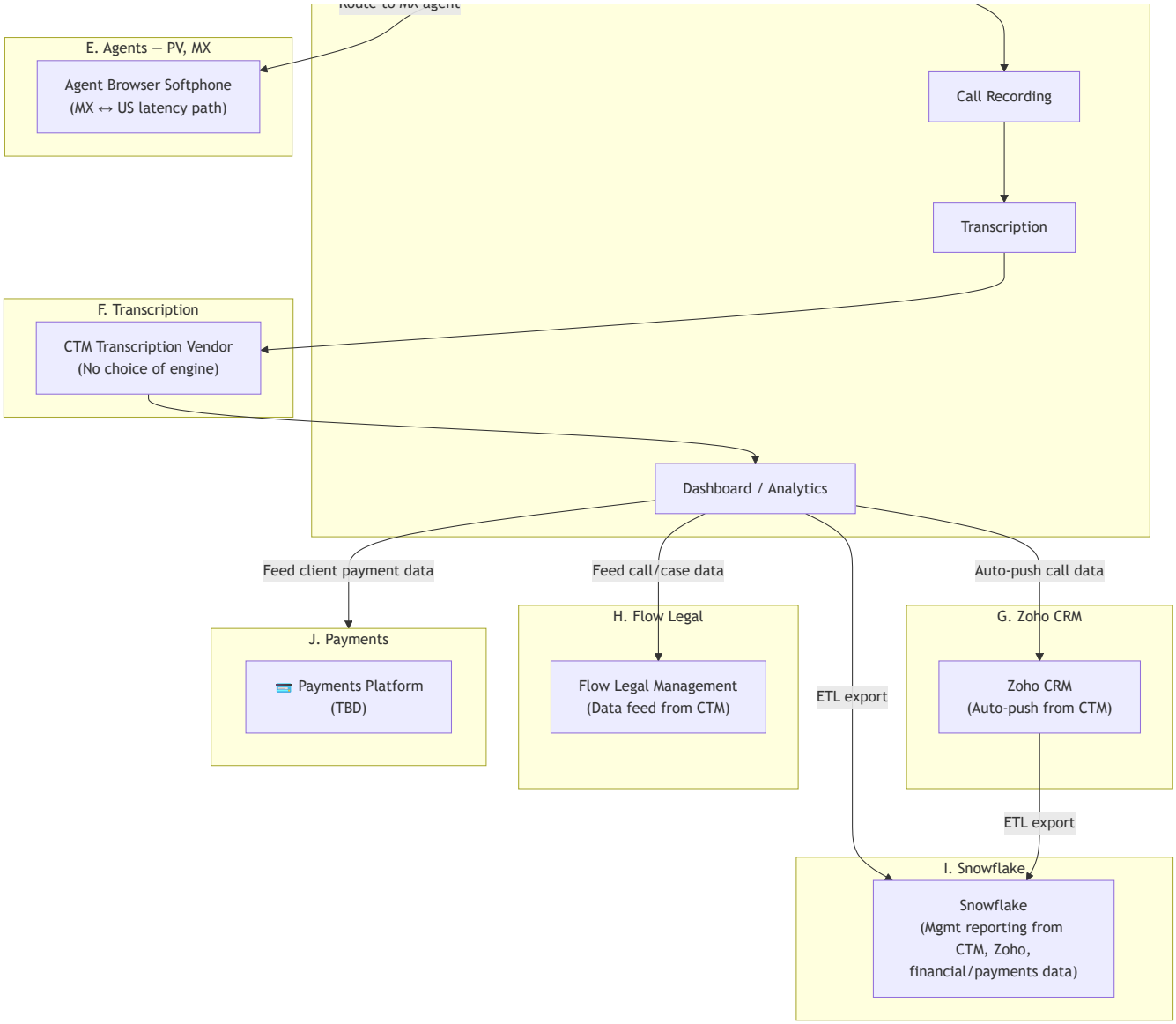
    - 4.3 Own the Call Path: Click-to-Call via WebRTC
    - 4.4 Own the Intelligence: Real-Time Transcription & AI Coaching
    - 4.5 Own the Numbers: Direct Telnyx Integration
    - 4.6 Own the CRM Feed: Direct Zoho Integration
    - 4.7 Own the Case Pipeline: Direct Flow Legal Integration
    - 4.8 Own the Analytics: Replace Snowflake with PostgreSQL
  - 5. Target Architecture Summary
    - Before (Legacy)
    - After (RustPBX Platform)
  - 6. Implementation Priority

---

1. Current Workflow Overview

The legacy call tracking architecture spans **ten layers (A–J)**, each owned by a different vendor or system. A customer's journey from Google Search through ad click, phone call, agent conversation, and downstream CRM/case management traverses all ten, accumulating latency, cost, and data fragmentation at every boundary.


A. Customer Device
Customer on Phone/Laptop
searches
returns ad results
clicks ad
B. Search & Ads
Google Search (Google Cloud)
Google Ads
Facebook Ads
Instagram Ads
click
click
click
C. CTM Browser-Side
Landing Page
CTM JS Tag
HTTP 100-300ms
CTM DNI API Call (Dynamic Number Insertion)
Browser Cookie (client-side session ↔ tracking number)
D. PSTN Network
Customer Dials DNI Number
PSTN / SS7
D. Twilio SIP
PSTN Gateway
Media Transcode
SIP Trunk
SIP INVITE
C. CTM Server-Side
Call Router
Route to MX agent



Vendor dependencies: 9+ external platforms (Twilio, CTM, Zoho, Flow Legal, Snowflake, AWS, Azure, GCC)

Billing redundancies: 3 separate per-call cost centers (ad spend, Twilio per-minute, CTM per-minute + platform fee)

Data silos: 7+ disconnected systems (each ad platform, Twilio logs, CTM analytics, Zoho CRM, Flow Legal, Snowflake, separate form tools)

Integration hub: CTM is the single point of failure — it feeds Zoho CRM, Flow Legal, and Snowflake. Removing CTM requires replacing all downstream integrations.

Reporting layer: Snowflake aggregates data from CTM and Zoho plus financial/payments data for management reporting. Both CTM and Zoho ETL pipelines must be redirected before Snowflake (or its replacement) can produce accurate reports.

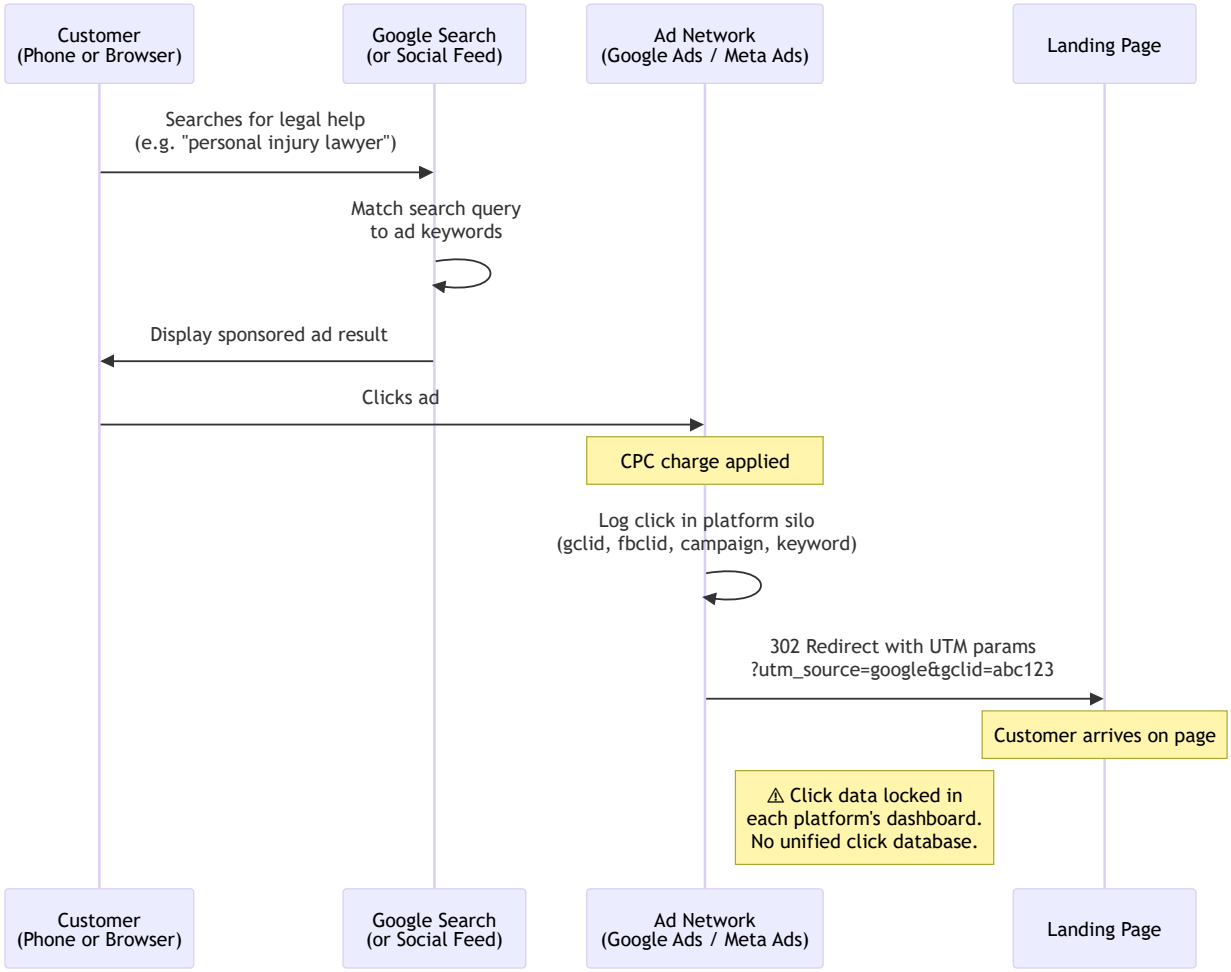
Geographic consideration: Agents operate from Puerto Vallarta, Mexico — all SIP media and signaling traverses the US-Mexico internet path, adding latency and introducing internet reliability concerns.

---

2. Step-by-Step Legacy Flow

Step 1: Customer Searches and Clicks an Ad (Layers A–B)

A prospective customer — on their phone or desktop browser — performs a **Google Search** (or browses Facebook/Instagram) and sees a paid advertisement. They click through to our landing page.



What happens technically:

- Customer initiates a **Google Search** query (or browses a social media feed) on their phone or desktop browser
- Google's ad auction matches the search query to our keyword bids and displays a sponsored result
- Customer clicks the ad; the ad platform registers a click event and charges our ad account (CPC or CPM model)
- Customer's browser is redirected to our landing page URL with UTM parameters and/or platform-specific click identifiers (gclid, fbclid, etc.) appended as query strings
- The ad platform logs the click in its own analytics silo (Google Ads dashboard, Meta Ads Manager, etc.)

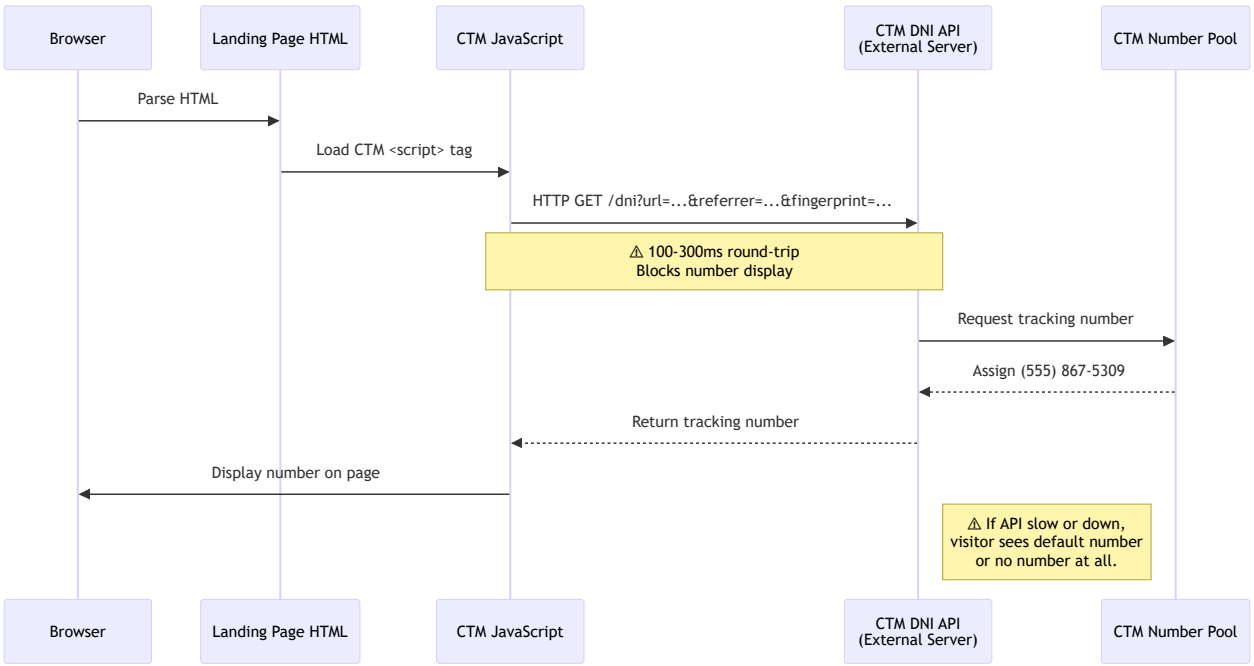
Data generated: Search query (Google only), click ID, campaign ID, ad group, keyword match type, placement (social), timestamp, device type (phone/desktop/tablet), geographic IP, referrer URL.

Where data lives: Locked inside each ad platform's proprietary dashboard. No unified click-level data store under our control. Google Search query data requires separate Search Console or Google Ads API

access.

Step 2: Landing Page Loads CTM JavaScript (Layer C)

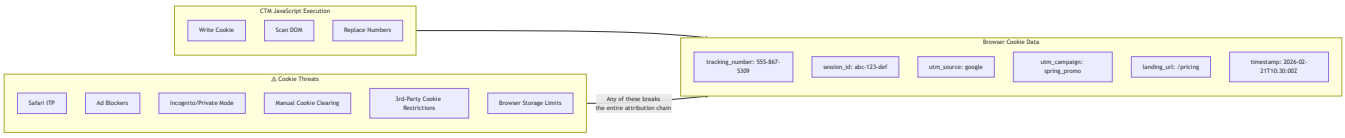
The landing page HTML includes a CTM JavaScript tracking tag that fires on page load.



Latency introduced: 100–300ms for the external API round-trip to CTM servers. This blocks the phone number from appearing on the page until the response returns. On slow connections or CTM API degradation, the visitor may see the default number or no number at all.

Step 3: Cookie Storage and Number Swap (Layer C)

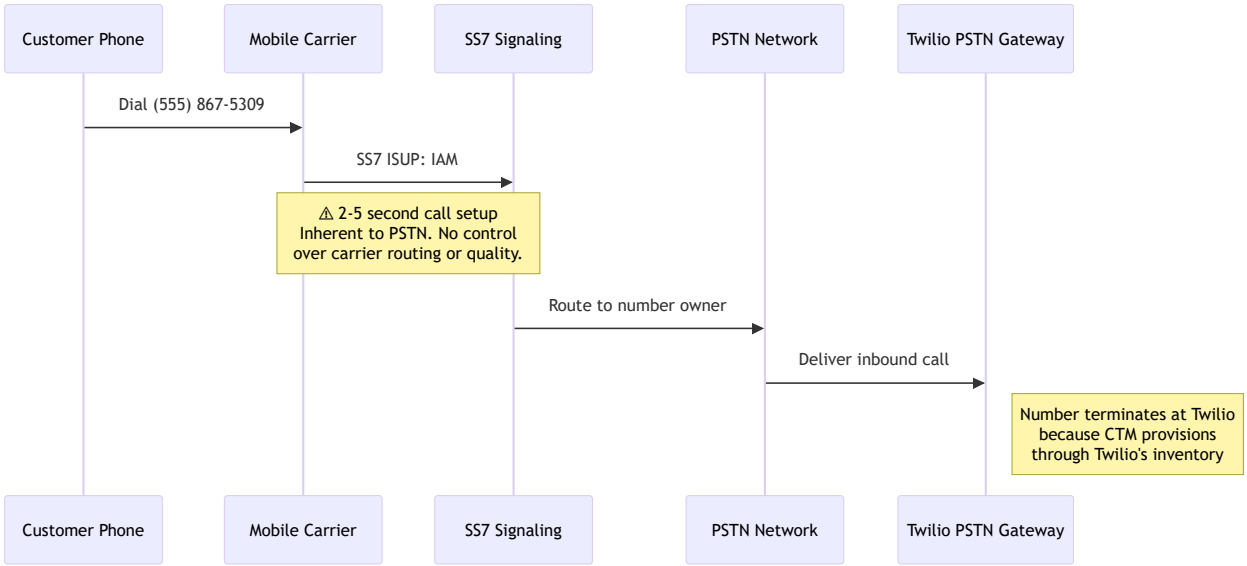
The CTM JavaScript stores tracking data in the visitor's browser and performs the number swap.



Fragility: The entire attribution chain depends on this cookie surviving. Browser cookie clearing, private/incognito mode, ITP (Safari Intelligent Tracking Prevention), third-party cookie restrictions, and ad blockers can all break the chain — silently. There is no fallback attribution mechanism.

Step 4: Customer Calls the Tracking Number (Layer D)

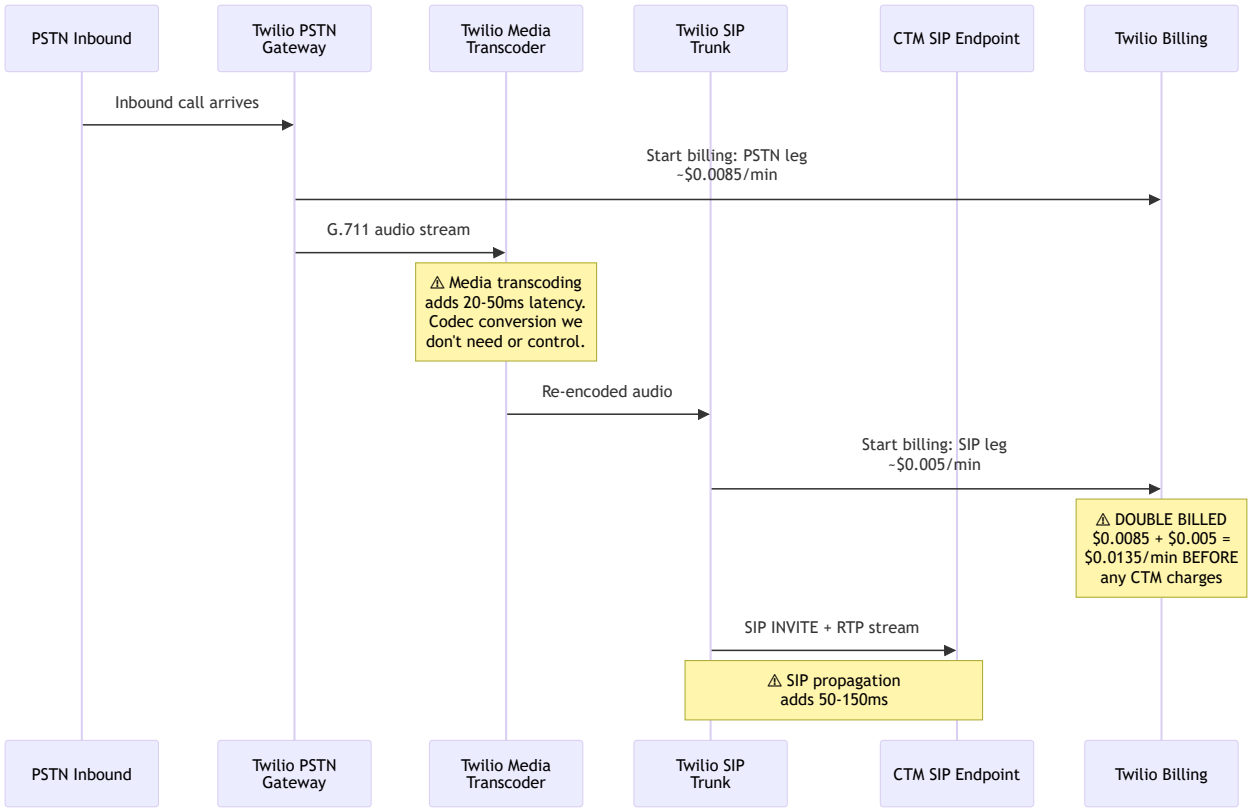
The customer sees the dynamically inserted phone number and dials it from their phone.



Latency introduced: Standard PSTN call setup (SS7 signaling) adds 2–5 seconds before the first ring. This is inherent to the telephone network and applies to any PSTN-based architecture. There is no control over carrier routing quality or path selection.

Step 5: Twilio Receives and Trunks the Call (Layer D)

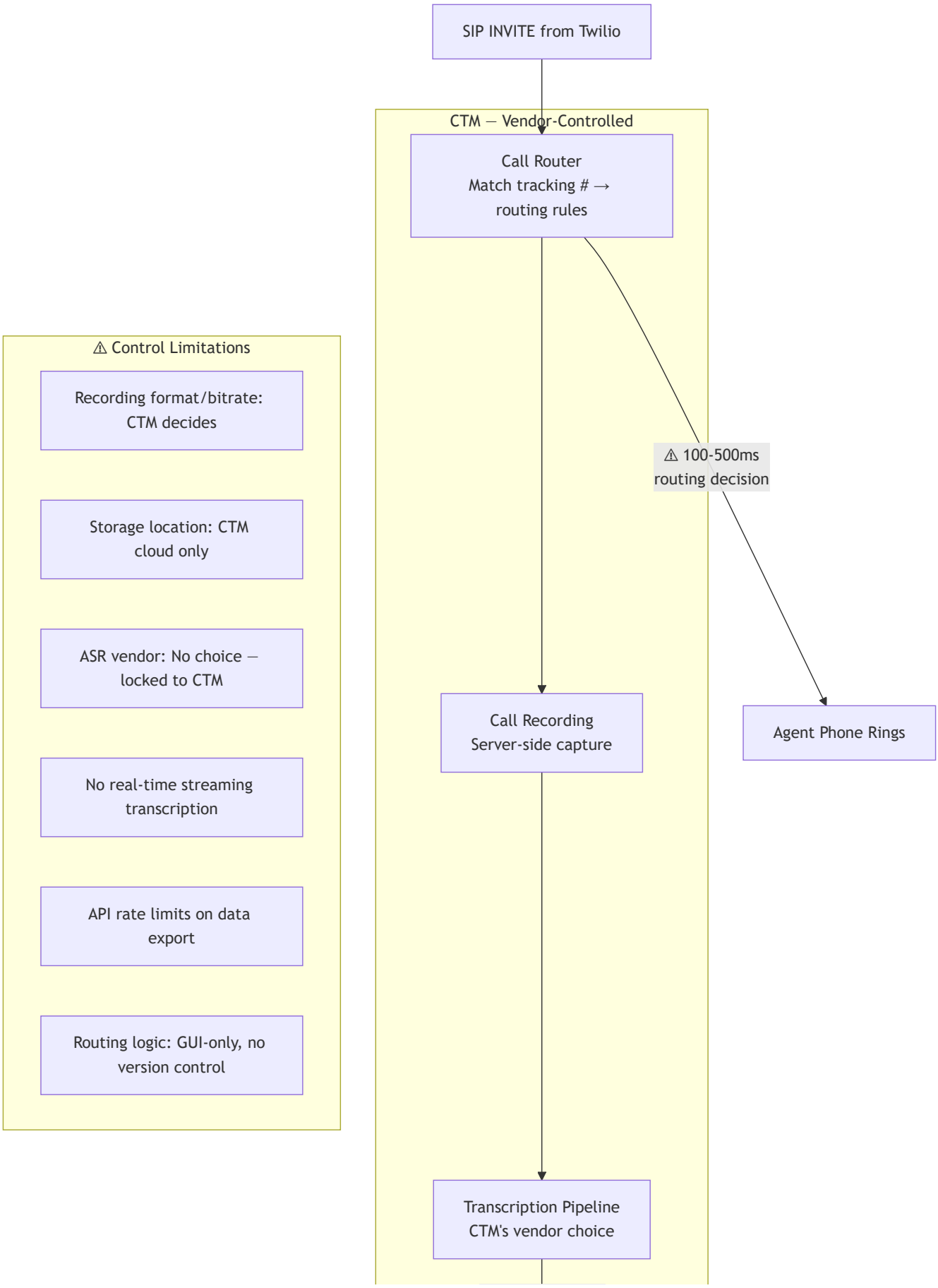
Twilio's PSTN gateway receives the inbound call and bridges it to CTM via SIP.

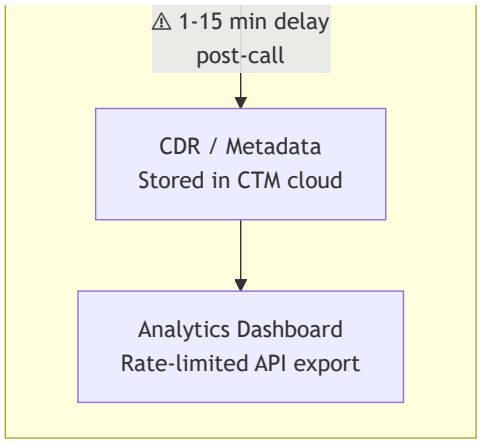


Cost impact: Twilio charges for both the inbound PSTN termination AND the outbound SIP trunk — you pay twice for the same call. Current Twilio rates: ~$0.0085/min inbound + ~$0.005/min SIP = ~$0.0135/min combined before any CTM charges.

Step 6: CTM Routes, Records, and Transcribes (Layer C)

CTM's platform receives the SIP INVITE and handles call management.


SIP INVITE from Twilio
CTM — Vendor-Controlled
Call Router
Match tracking # →
routing rules
⚠ 100-500ms
routing decision
Agent Phone Rings
Call Recording
Server-side capture
Transcription Pipeline
CTM's vendor choice
⚠ Control Limitations
Recording format/bitrate:
CTM decides
Storage location: CTM
cloud only
ASR vendor: No choice —
locked to CTM
No real-time streaming
transcription
API rate limits on data
export
Routing logic: GUI-only, no
version control



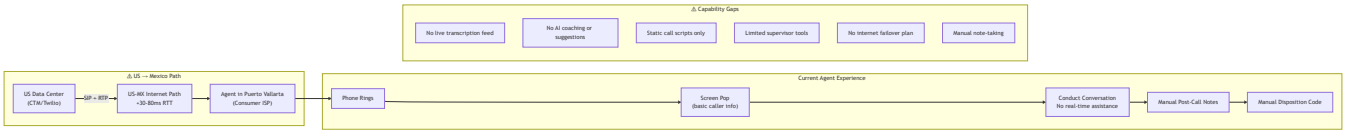
Latency introduced: CTM's routing decision adds 100–500ms before the agent's phone rings. Transcription is typically available 1–15 minutes post-call depending on duration and queue depth.

Control limitations:

- Recording format, bitrate, and storage location are CTM's decisions, not ours
- Transcription vendor/model is whatever CTM has contracted — no ability to choose Deepgram vs. Groq vs. OpenAI Whisper based on our accuracy/cost preferences
- No real-time transcription streaming during live calls (post-call only on most tiers)
- API rate limits constrain data export for custom analytics
- Call data lives in CTM's cloud — we don't own the storage

Step 7: Agent Handles the Call (Layer E — Puerto Vallarta, Mexico)

The agent — located in **Puerto Vallarta, Mexico** — receives the call via browser softphone or desktop SIP client. All SIP signaling and RTP media traverse the US-Mexico internet path, adding latency and introducing reliability concerns.



Geographic and reliability concerns:

- SIP signaling from CTM/Twilio US data centers to Puerto Vallarta adds **30-80ms round-trip** latency on top of PSTN call setup time
- Agent relies on **consumer-grade Mexican ISP** — no dedicated circuit or failover
- Internet outages in PV mean agents go completely offline with no automatic rerouting

- RTP media quality degrades if the US-MX path experiences jitter or packet loss
- No codec optimization for the international path (CTM/Twilio control codec selection)

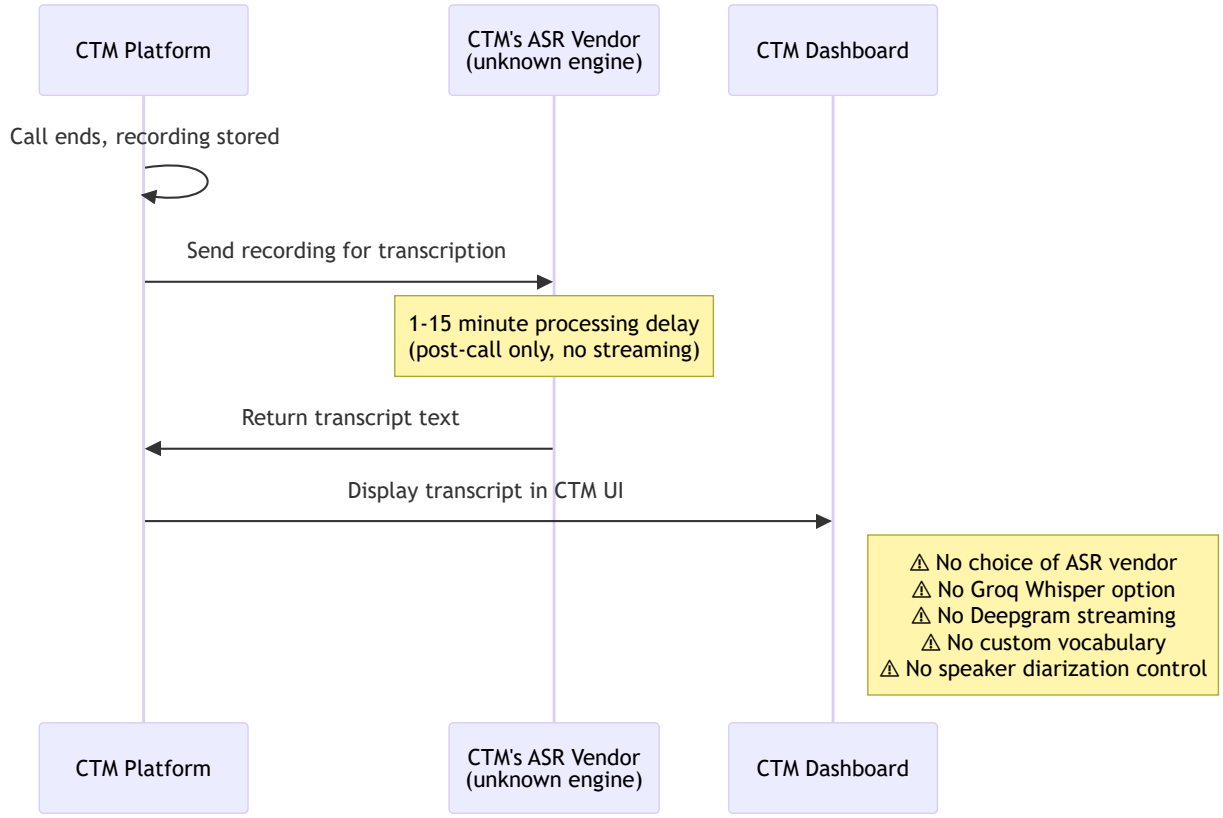
Capability gaps:

- No live transcription feed visible to the agent during the call
- No real-time AI coaching or suggested responses
- Call scripts are static documents, not dynamically adapted to the conversation
- Supervisor monitoring is limited to CTM's barge/whisper features
- Post-call review is the only analysis option — no in-call intervention
- No failover strategy if Puerto Vallarta internet drops mid-call

---

Step 8: CTM Transcribes the Call (Layer F — Vendor-Locked ASR)

After the call ends, CTM sends the recording to its own transcription vendor. We have **no choice** of ASR engine, model, or pricing.



What we cannot control today:

- Which ASR model processes our recordings (CTM chooses the vendor)
- Transcription latency — always post-call, typically 1-15 minutes after call ends
- No real-time streaming transcription during live calls
- No custom vocabulary for legal/medical terminology our callers use
- No speaker diarization control (separating agent vs. caller)
- Transcript accuracy and language support are whatever CTM's vendor provides

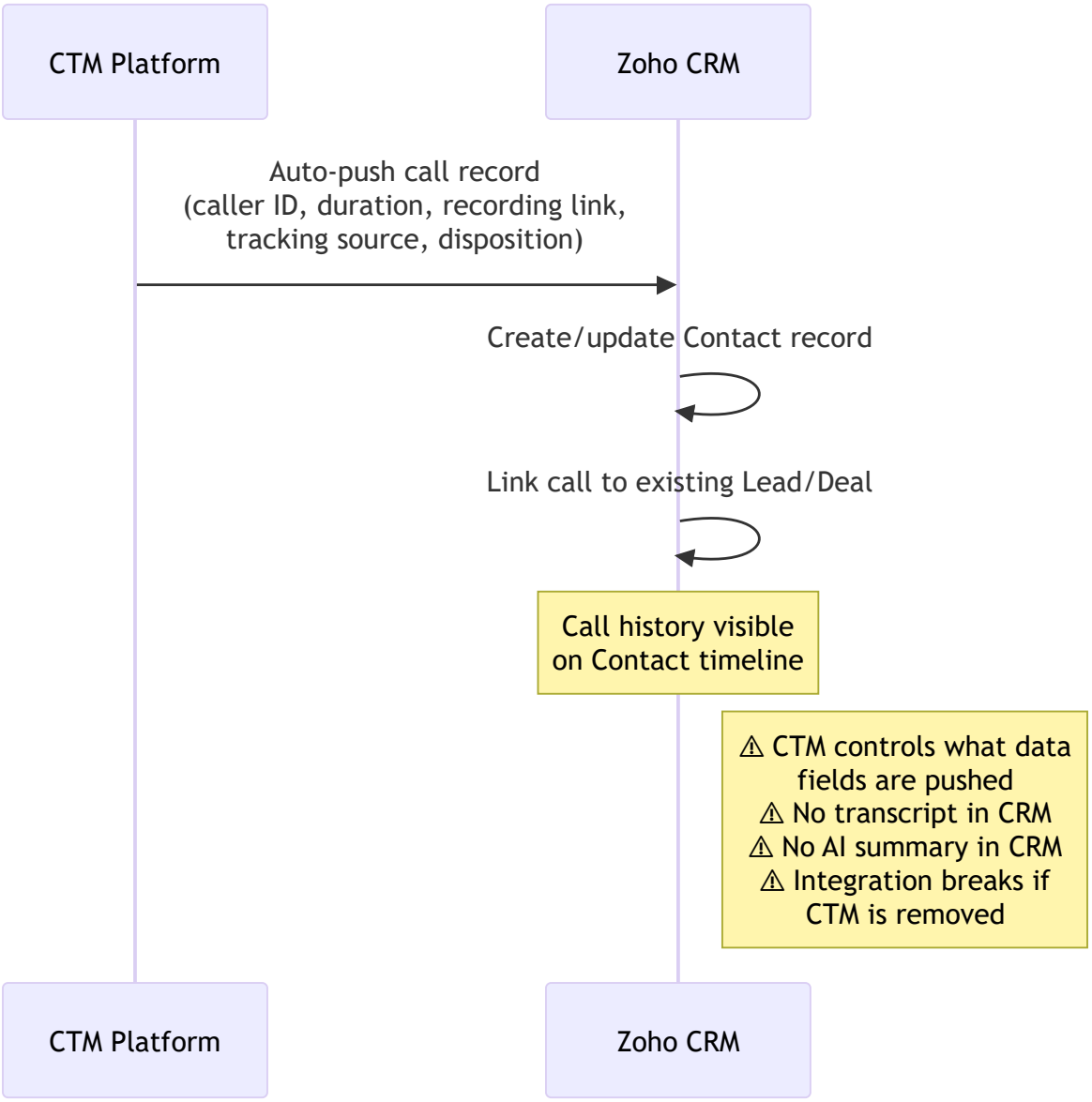
What we want (to be detailed in ARCHITECTURE_VISION.md):

- **Groq Whisper Large v3 Turbo** for batch post-call transcription at $0.04/hour
- **Deepgram Nova-3** for real-time streaming transcription during live calls
- Full control over custom vocabulary, diarization, and model selection

---

Step 9: CTM Pushes Data to Zoho CRM (Layer G)

CTM automatically pushes call data to **Zoho CRM** after each call. This is our CRM system of record for customer contacts and sales pipeline.



What gets pushed today:

- Caller phone number, call duration, timestamp
- Call recording link (points back to CTM-hosted recording)
- Tracking source / campaign attribution from DNI
- Agent who handled the call, disposition code

What doesn't get pushed:

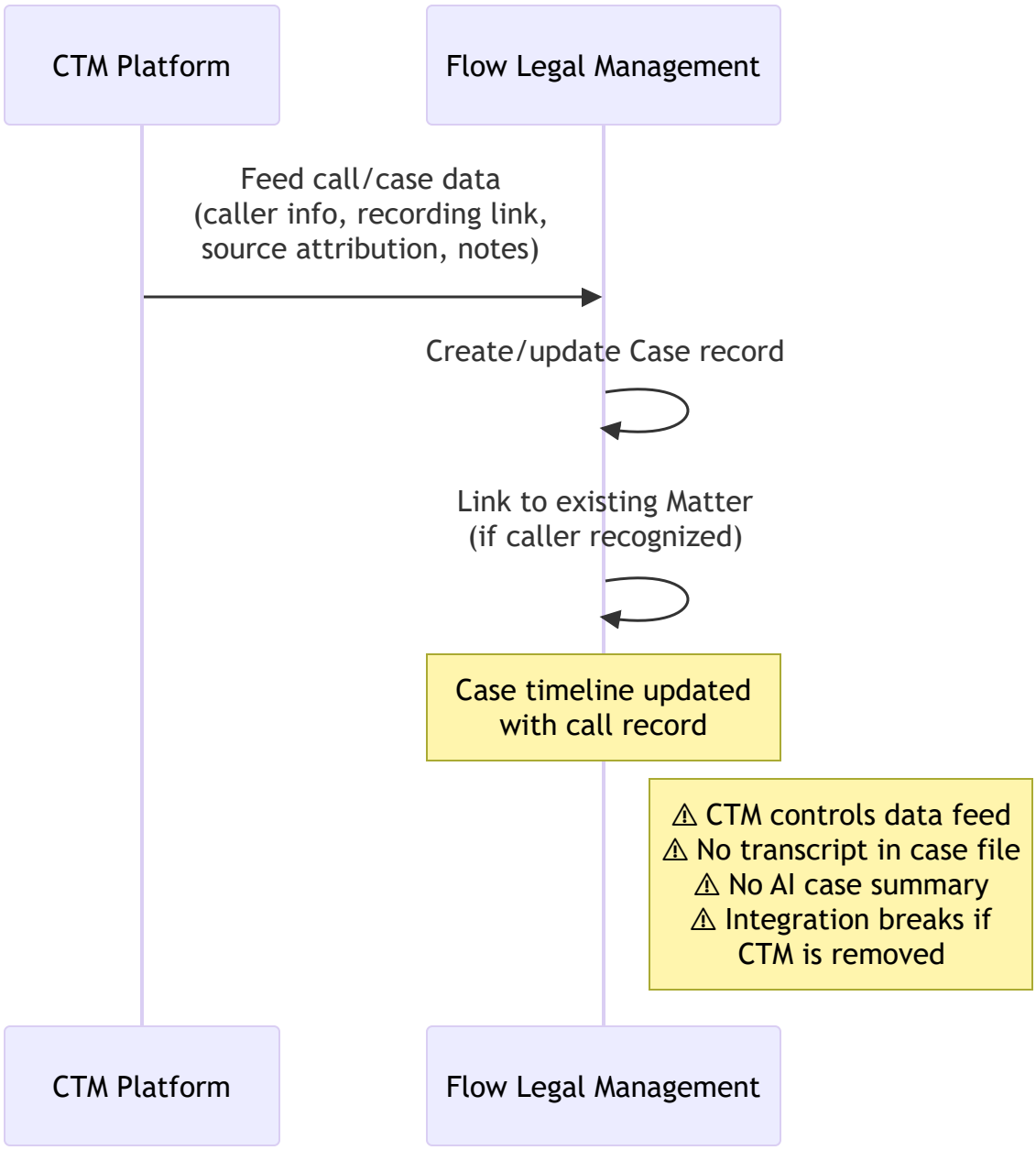
- Full transcript text (too large for standard CRM push)
- AI-generated call summary or action items
- Sentiment analysis or compliance flags

- Form submission data correlated with the call

Integration dependency: Zoho CRM integration is **entirely dependent on CTM** as the middleware. When we remove CTM, RustPBX must replicate this data push via Zoho's Telephony API or REST API.

Step 10: CTM Feeds Data to Flow Legal Management (Layer H)

CTM feeds call and case data to **Flow Legal Management**, our case management system for tracking legal matters tied to incoming calls.



What gets fed today:

- Caller information and call metadata
- Call recording link (CTM-hosted)
- Source attribution from call tracking
- Agent notes and disposition

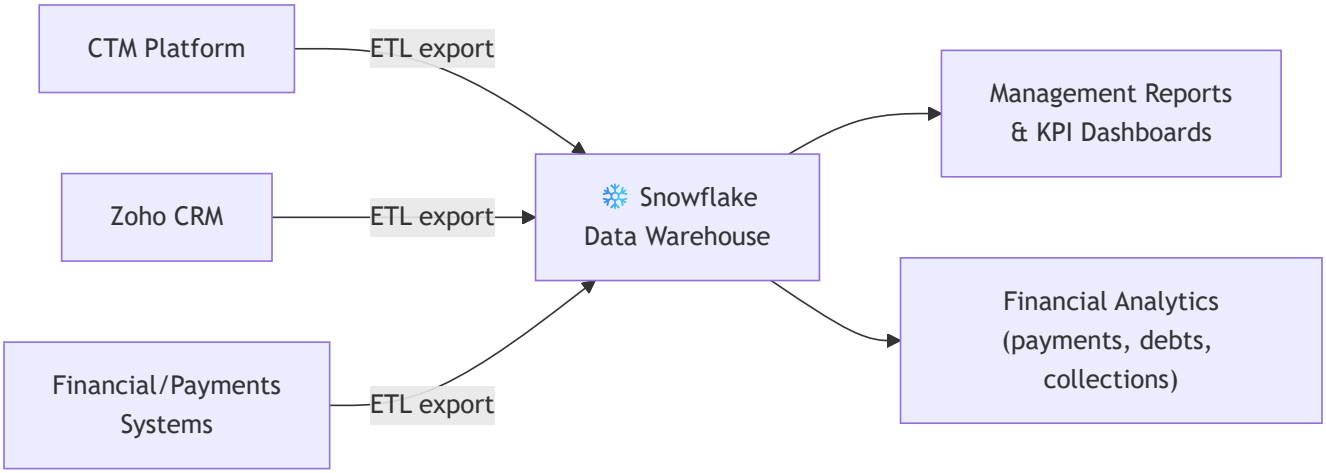
What doesn't get fed:

- Searchable transcript text for case discovery
- AI-generated case summary or intake notes
- Compliance flags (HIPAA/TCPA mentions detected during call)
- Linked form submissions from the same caller session

Integration dependency: Like Zoho, the Flow Legal integration runs through CTM. Removing CTM requires building a direct RustPBX → Flow Legal data pipeline, likely via Flow Legal's API or webhook integration.

Step 11: CTM and Zoho Feed Snowflake for Management Reporting (Layer I)

Data from CTM and Zoho CRM is exported (ETL) into **Snowflake**, a cloud data warehouse used for cross-system management reporting and financial analytics.



What Snowflake enables today:

- Cross-system queries (ad attribution → call → case → payment full-funnel ROI)
- Call center performance reporting (volume, handle time, disposition rates)

- Marketing attribution analytics (cost per lead by campaign, conversion rates)
- Financial reporting (revenue per case type, outstanding debt aging, collection rates)

What doesn't work well:

- Reports are only as current as the last ETL run — no real-time dashboards
- Data quality depends on CTM and Zoho exports being complete and consistent
- Adding new metrics requires modifying ETL pipelines across multiple source systems
- Three separate billing relationships (Snowflake compute, CTM data export, Zoho API)

Data dependency: Snowflake queries depend on data from CTM and Zoho. Replacing either upstream system requires redirecting the corresponding ETL pipeline, or Snowflake reports will go stale or break.

---

3. Inefficiency Analysis by Layer

3.1 Ad Click Attribution (Layer B)

Inefficiency	Impact	Severity
Click data siloed per ad platform	No unified funnel view across Google, Meta, etc.	High
No server-side click logging	Dependent on client-side JS + cookies for attribution	High
UTM parameters are fragile	Users can modify URLs, parameters get stripped by redirects	Medium
No first-party click database	Cannot correlate clicks → calls → outcomes in one query	High
Platform-reported conversions lag	Google/Meta conversion APIs have 24-72hr attribution delays	Medium

3.2 Browser-Side Tracking (Layer C)

Inefficiency	Impact	Severity
External API dependency for DNI	100-300ms page load penalty per visit	High
Cookie-based tracking chain	Breaks under ITP, ad blockers, private browsing	Critical
No form capture integration	Separate tools needed for lead forms vs. call tracking	Medium
Single-vendor number pool	CTM controls number inventory, porting, and pricing	High
No offline conversion bridge	Clicks that lead to in-store visits have no tracking path	Low

3.3 PSTN Call Path (Layer D)

Inefficiency	Impact	Severity
2-5 second call setup latency	Inherent to PSTN, unavoidable for traditional calls	Medium
No internet-only call option	Every call must traverse PSTN even if both parties have broadband	High
Per-minute carrier charges	Applies to every call regardless of internal routing capability	Medium
No video call option	PSTN is audio-only by definition	Medium

3.4 Twilio SIP Trunking (Layer D)

Inefficiency	Impact	Severity
Double billing (PSTN + SIP legs)	~$0.0135/min combined cost before CTM fees	High
Unnecessary media transcoding	20-50ms latency for codec conversion we don't need	Medium
Vendor lock-in	Number porting from Twilio requires 2-4 week lead time	High
No codec negotiation control	Cannot optimize for Opus or wideband audio	Medium
Geographic routing opacity	No visibility into Twilio's internal call path decisions	Low

3.5 CTM Platform (Layer C)

Inefficiency	Impact	Severity
Second per-minute fee layer	CTM charges on top of Twilio charges	Critical
No real-time transcription	Post-call only on most plan tiers	High
Locked recording storage	Cannot choose format, retention, or storage location	High
Transcription vendor locked	No ability to benchmark or swap ASR providers	High
API rate limits on export	Constrains custom analytics and data warehousing	Medium
No agent coaching hooks	Platform has no LLM integration points	High
Routing logic is GUI-only	Cannot version-control or programmatically manage routes	Medium

3.6 Agent Experience (Layer E)

Inefficiency	Impact	Severity
No live transcription	Agent cannot see real-time text of conversation	High
No AI-assisted coaching	No suggested responses, objection handling, or compliance alerts	High
Static call scripts	Scripts don’t adapt to conversation context	Medium
Limited supervisor tools	Barge/whisper only, no real-time dashboard of all active calls	Medium
Manual post-call work	Agent must hand-type notes and disposition codes	Medium

3.7 Agent Geographic Location (Layer E)

Inefficiency	Impact	Severity
US-MX internet path adds 30-80ms RTT	Voice quality degradation on international SIP path	High
Consumer-grade ISP in Puerto Vallarta	No dedicated circuit, no SLA for uptime or bandwidth	Critical
No internet failover strategy	ISP outage = agents completely offline, no automatic rerouting	Critical
No codec optimization for international path	CTM/Twilio control codec selection, cannot optimize for jitter	Medium
Jitter/packet loss on cross-border path	RTP media quality degrades unpredictably on consumer ISP	High
No regional SIP proxy	All signaling routes through US data center, no MX-local optimization	Medium

3.8 Transcription (Layer F)

Inefficiency	Impact	Severity
No choice of ASR engine	Cannot use Groq Whisper Turbo ($0.04/hr) or Deepgram Nova-3	High
Post-call only transcription	1-15 minute delay, no real-time streaming during live calls	High
No custom vocabulary	Cannot optimize for legal/medical terminology callers use	Medium
No speaker diarization control	Cannot reliably separate agent vs. caller in transcripts	Medium
Transcript locked in CTM	Cannot export/search transcripts outside CTM's dashboard	High
No transcript push to CRM/case mgmt	Zoho and Flow Legal don't receive searchable transcript text	High

3.9 CRM Integration (Layer G — Zoho via CTM)

Inefficiency	Impact	Severity
CTM is the sole integration middleware	Removing CTM breaks Zoho call data feed completely	Critical
No transcript text in CRM records	Zoho contact timeline has call metadata but not transcript	High
No AI summary in CRM	Agent must manually write call notes; no auto-generated summary	High
Recording links point to CTM-hosted files	If CTM account closes, all recording links in Zoho break	High
CTM controls pushed field mapping	Cannot customize which data fields Zoho receives	Medium
No form-to-call correlation in CRM	Web form submissions and related calls are not linked in Zoho	Medium

3.10 Case Management (Layer H — Flow Legal via CTM)

Inefficiency	Impact	Severity
CTM is the sole data feed source	Removing CTM breaks Flow Legal call/case data pipeline	Critical
No searchable transcript in case files	Legal case discovery cannot search call transcript text	High
No AI-generated intake summary	Agents manually create intake notes instead of auto-generated	High
Recording links depend on CTM hosting	CTM account closure orphans all recordings linked in cases	High
No compliance flag automation	HIPAA/TCPA/PCI mentions in calls not auto-flagged in case files	Medium
No form submission linkage	Web form submissions not correlated with related case calls	Medium

3.11 Management Reporting (Layer I — Snowflake)

Inefficiency	Impact	Severity
Reports lag behind reality (batch ETL)	Decisions made on stale data; no real-time dashboards	High
Depends on CTM + Zoho ETL pipelines	Replacing either upstream system breaks Snowflake reports	Critical
Three separate vendor billing layers	Snowflake compute + CTM data export + Zoho API costs compound	Medium
Adding new metrics requires multi-system ETL changes	Weeks of work to surface a new cross-system KPI	High
Financial/debt data pipeline is separate from call data	No unified view linking call activity to payment outcomes	High
No AI-enriched data in warehouse	Transcripts, sentiment, compliance flags not available for analytics	High

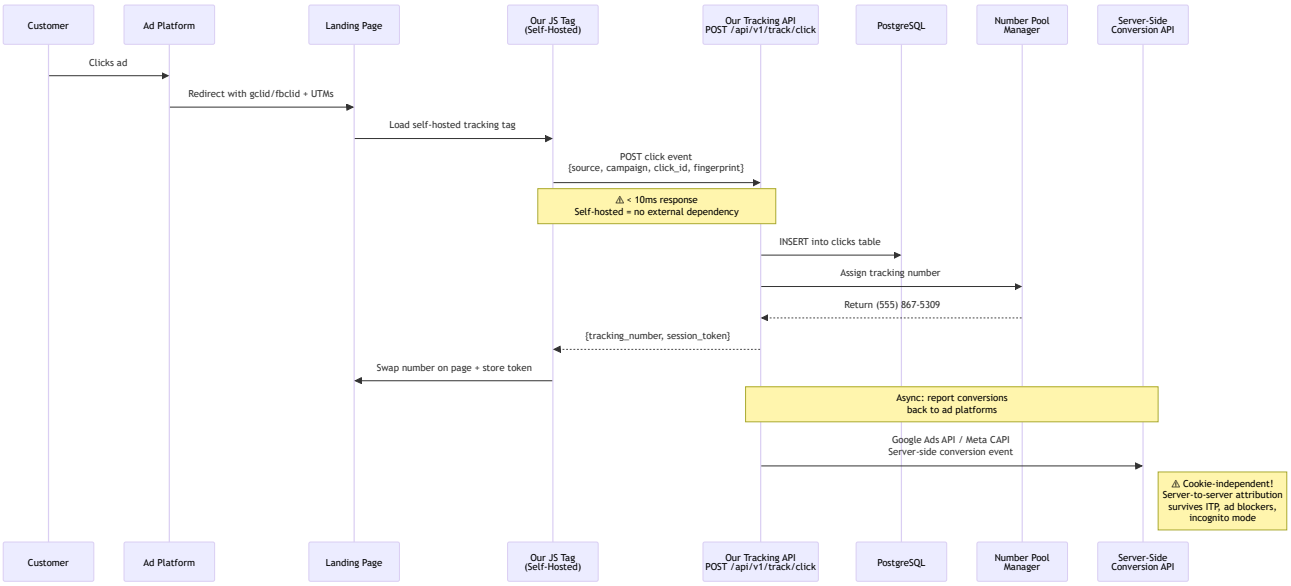
---

4. Redesign Opportunities

4.1 Own the Click: First-Party Ad Tracking API

Current state: Ad clicks are logged only in each ad platform's dashboard. Our landing page relies on CTM's JavaScript to capture attribution. We have no unified click database.

Opportunity: Build a lightweight first-party tracking API that our landing page calls on every click event, logging attribution data directly into our PostgreSQL database before any third-party dependency enters the picture.

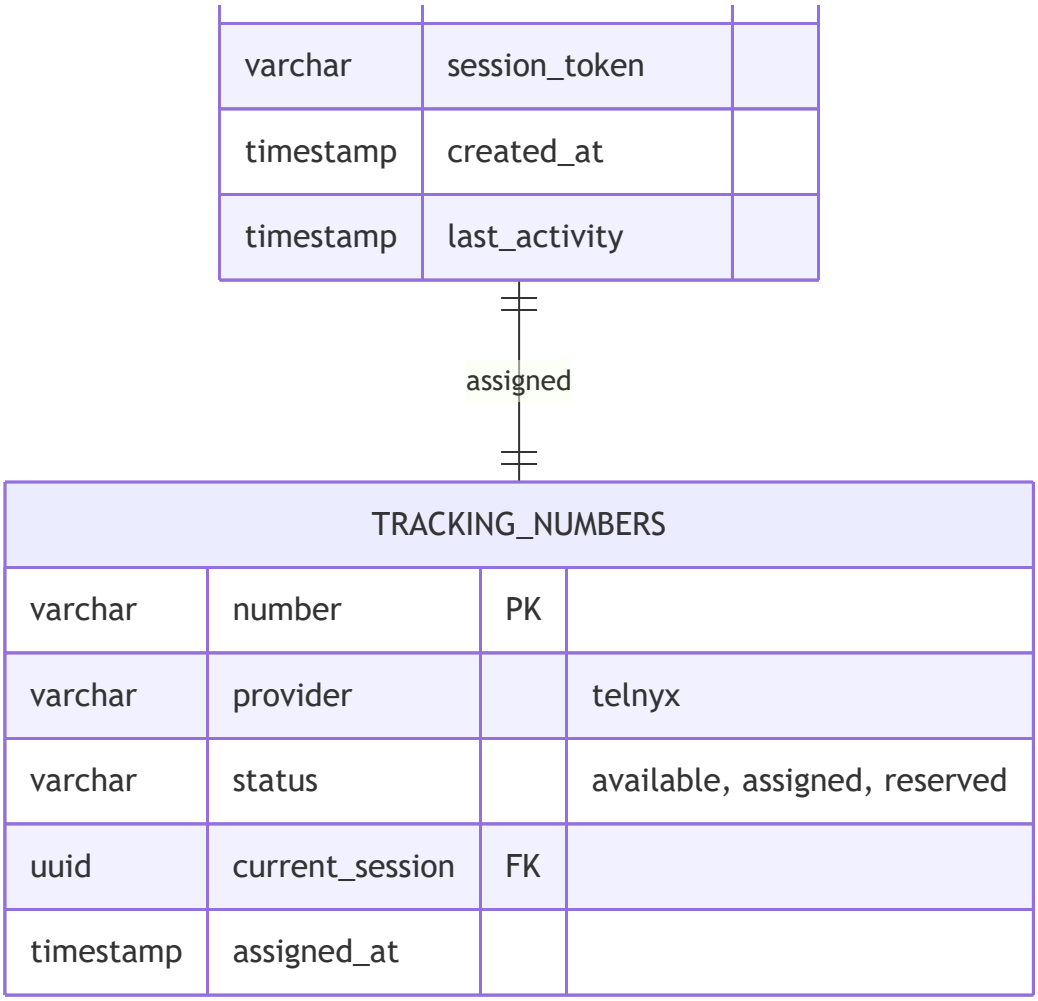


Database schema (clicks table):

CLICKS			
uuid	click_id	PK	
varchar	source		google, facebook, instagram
varchar	campaign_id		
varchar	ad_group		
varchar	keyword		Google Search only
varchar	gclid		Google click ID
varchar	fbclid		Meta click ID
varchar	device_type		mobile, desktop, tablet
inet	ip_address		
varchar	geo_region		
varchar	landing_url		
varchar	referrer_url		
uuid	session_id	FK	
timestamp	created_at		

belongs to

SESSIONS		
uuid	session_id	PK
varchar	tracking_number	FK
varchar	visitor_fingerprint	



Implementation approach:

- Lightweight REST endpoint: `POST /api/v1/track/click` accepts source, campaign, click ID, and visitor fingerprint. Returns a JSON response with the assigned tracking number and a session token.
- Our own JavaScript tag (replacing CTM's) calls this endpoint on page load. The tag is self-hosted — zero external API dependency, sub-10ms response time since it hits our own infrastructure.
- Server-side click logging with Google Ads API / Meta Conversions API integration for bidirectional attribution (we report conversions back to the ad platforms via their server-side APIs, eliminating client-side cookie dependency for conversion reporting).
- Unified clicks table in PostgreSQL that can be JOINed directly to call records, form submissions, and outcomes in a single query.

Benefits:

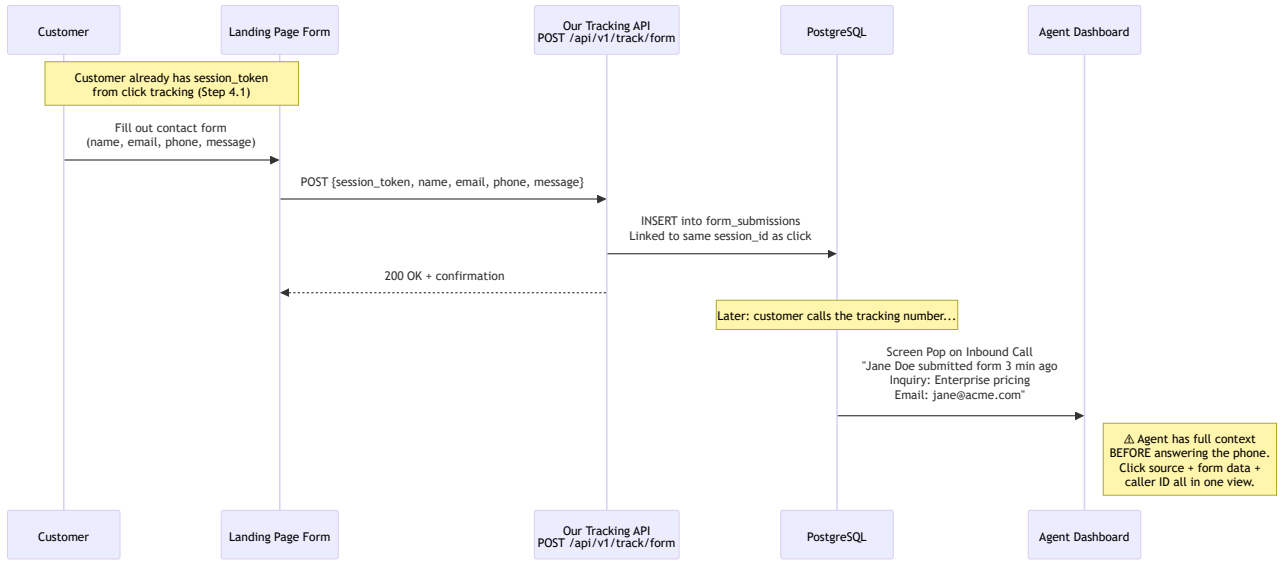
- Eliminate 100-300ms CTM DNI API latency
- Cookie-independent server-side attribution (session tokens + server-side ad platform APIs)
- Unified click → call → outcome analytics in one database

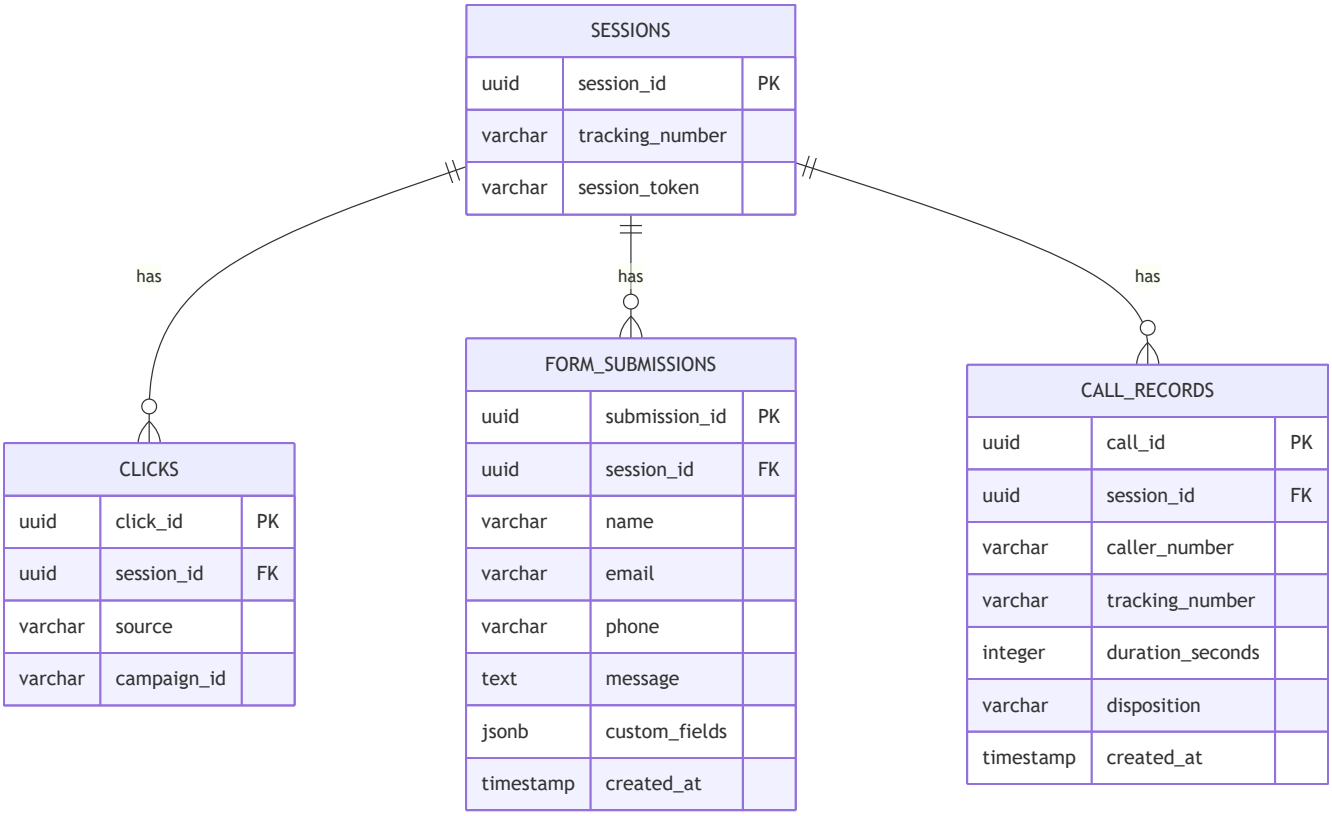
- Full control over number pool management and assignment logic
- No per-request API rate limits from third parties

4.2 Own the Form: Integrated Lead Capture

Current state: Form submissions on landing pages use separate tools (HubSpot, Typeform, Google Forms, etc.) that don't natively connect to call tracking data. Correlating "this person filled out a form AND called" requires manual data stitching across platforms.

Opportunity: Build form capture directly into the tracking platform so that form submissions and phone calls share the same session identity and attribution chain.





Implementation approach:

- Embeddable form component (Leptos WASM widget or lightweight JS) that submits to `POST /api/v1/track/form` with the session token from click tracking
- Form data stored in PostgreSQL with foreign key to the sessions/clicks table
- When the same visitor later calls, the agent gets a screen pop showing: "This caller submitted a form 3 minutes ago — Name: Jane Doe, Inquiry: Pricing for Enterprise plan"
- Form submissions can trigger automated actions: send confirmation email, create CRM contact, schedule callback, or initiate a proactive outbound call

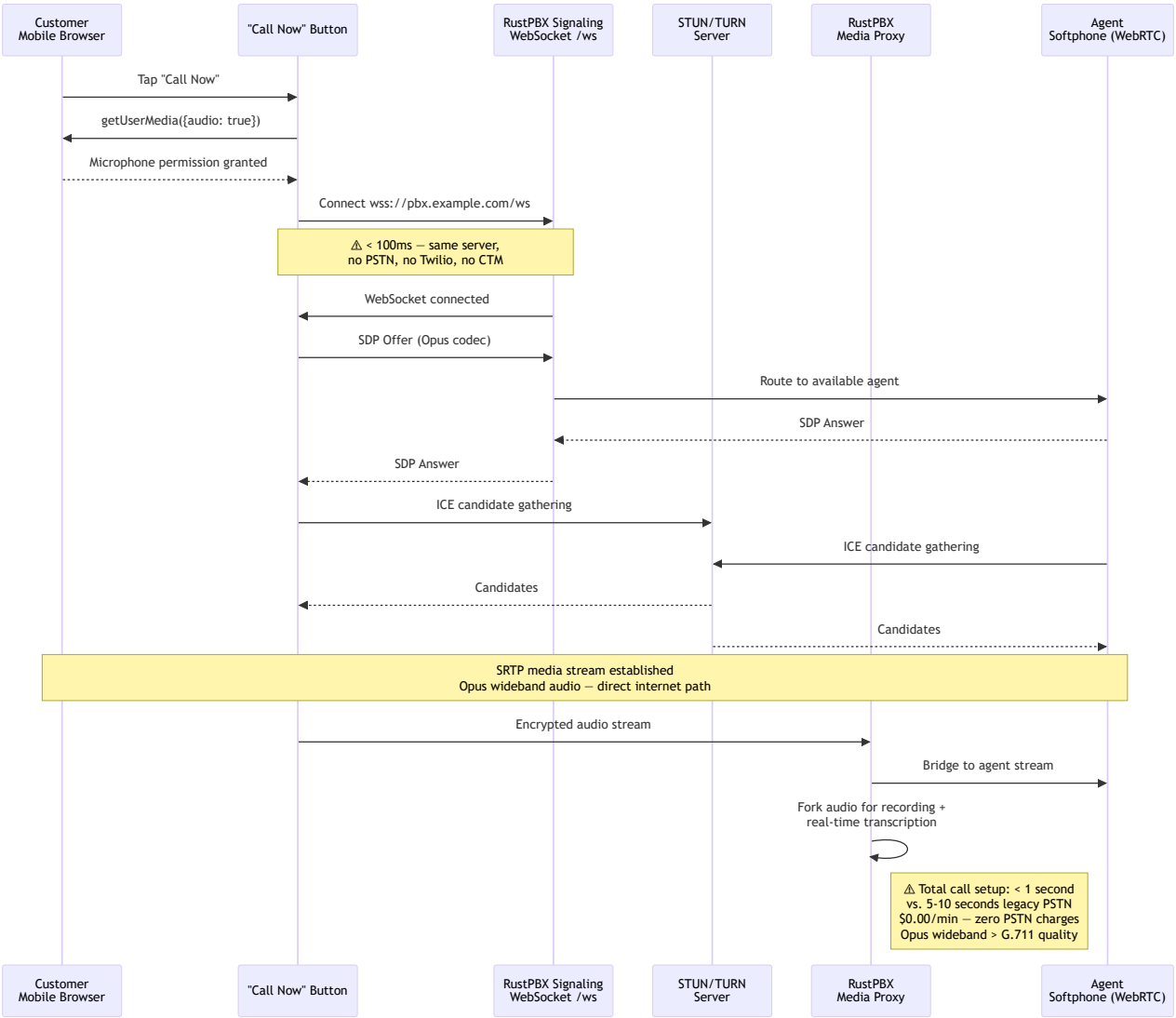
Benefits:

- Single session identity across clicks, forms, and calls
- Agent context before they even answer the phone
- No separate form tool subscription
- Form + call conversion data in one analytics query

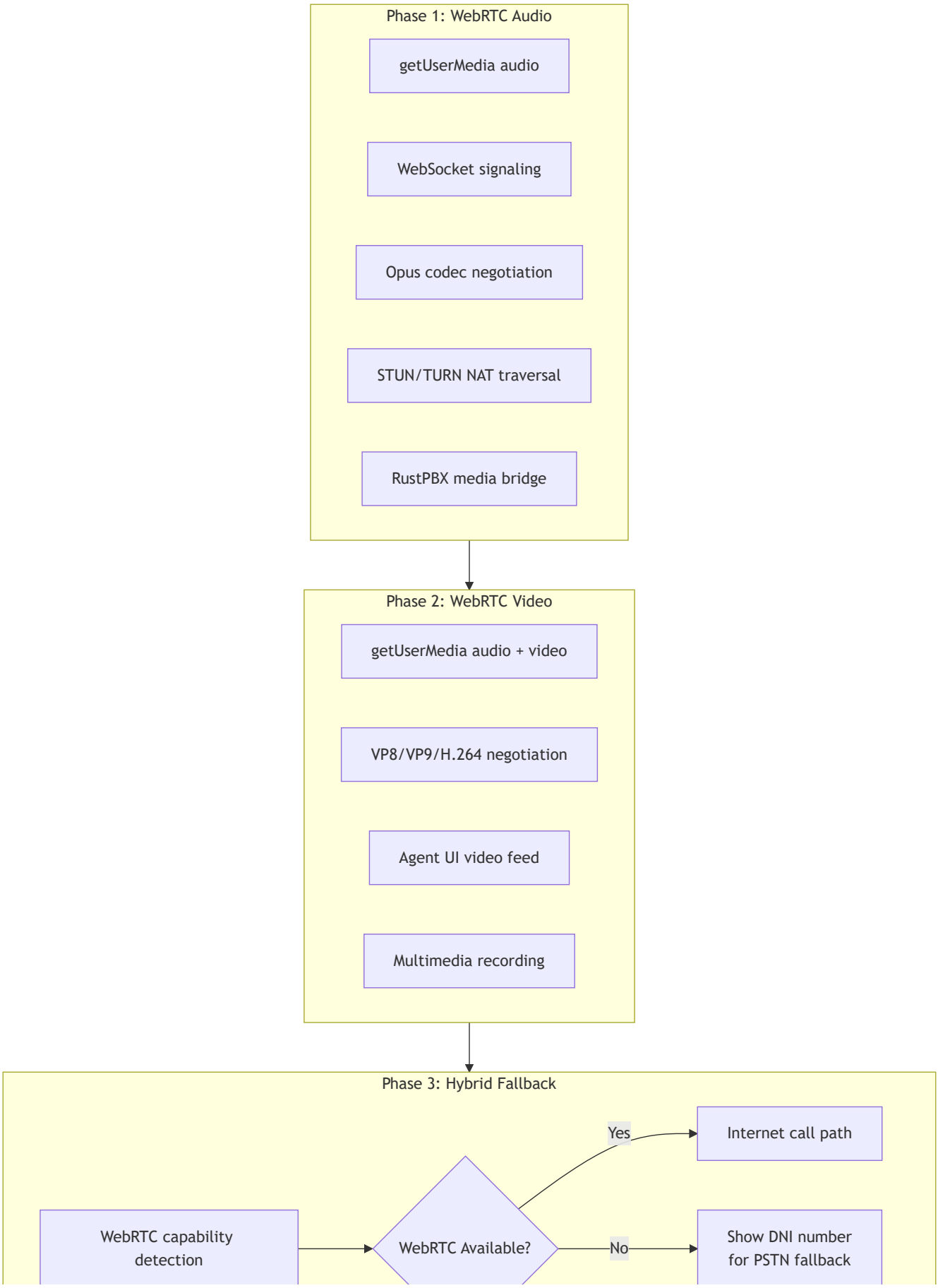
4.3 Own the Call Path: Click-to-Call via WebRTC

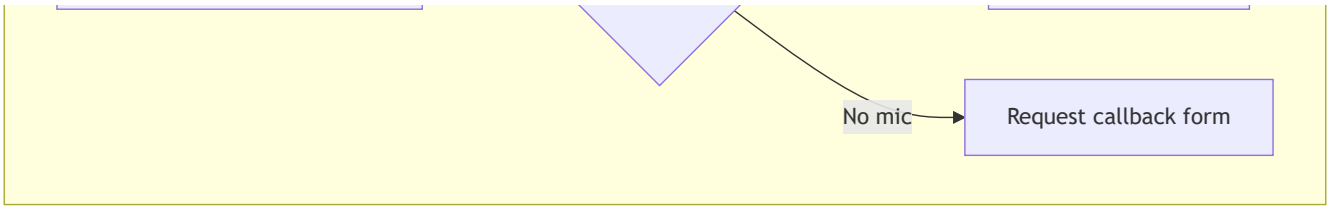
Current state: Every customer call must traverse the PSTN network, even when both the customer (on a smartphone with LTE/WiFi) and the agent (on a SIP softphone with broadband) have internet connectivity. This forces PSTN call setup latency, carrier charges, and audio quality limitations onto every interaction.

Opportunity: Offer a click-to-call button on landing pages that initiates a direct internet-based audio or video call from the customer's smartphone browser to our call center, completely bypassing PSTN.



Three implementation phases:


Phase 1: WebRTC Audio
getUserMedia audio
WebSocket signaling
Opus codec negotiation
STUN/TURN NAT traversal
RustPBX media bridge
Phase 2: WebRTC Video
getUserMedia audio + video
VP8/VP9/H.264 negotiation
Agent UI video feed
Multimedia recording
Phase 3: Hybrid Fallback
WebRTC capability detection
WebRTC Available?
Yes
Internet call path
No
Show DNI number for PSTN fallback



Benefits:

- Sub-second call setup (vs. 2-5 seconds for PSTN)
- Zero PSTN charges — internet-only media path
- Opus wideband audio quality (superior to G.711 PSTN)
- Video capability opens entirely new service modalities
- Customer stays on the landing page during the call (no app switch to phone dialer)
- Full call context preserved: we know exactly which page, ad, campaign, and session triggered the call because the same JavaScript that tracked the click initiates the call

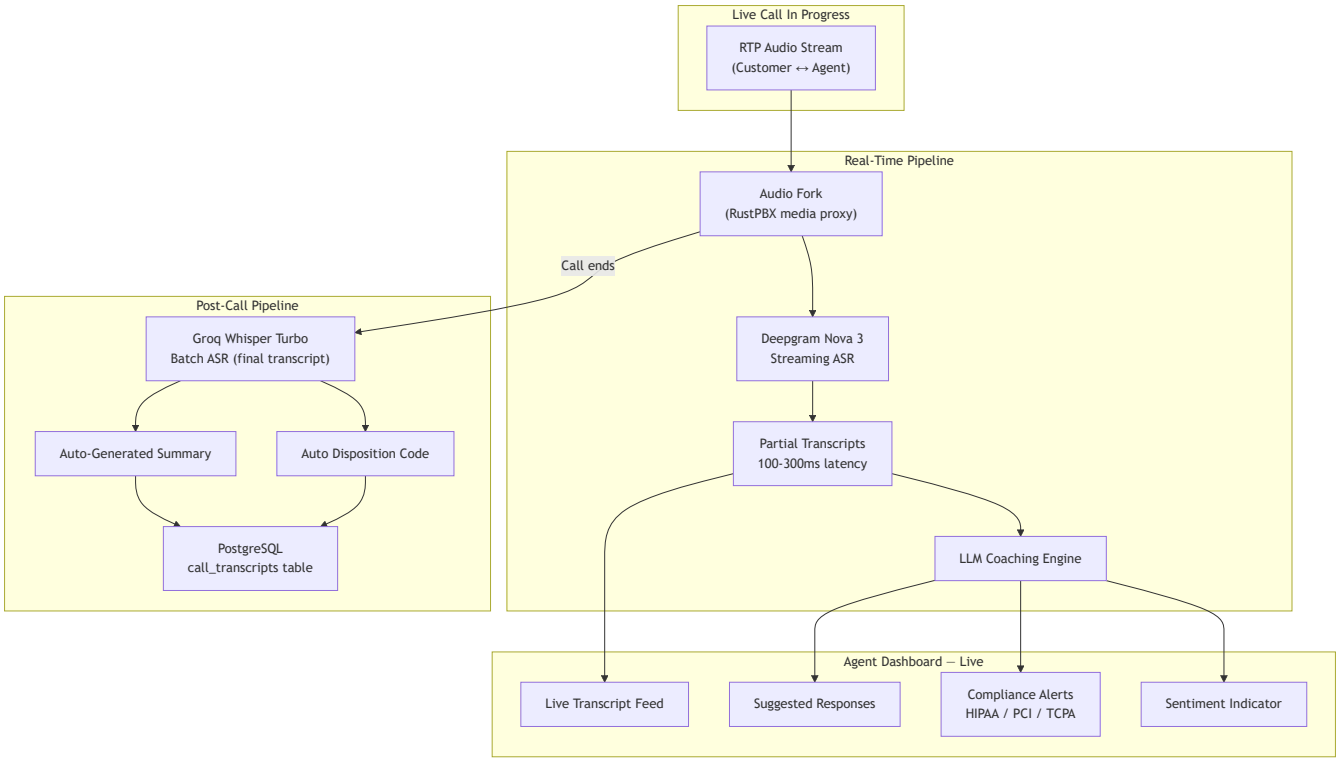
Technical requirements:

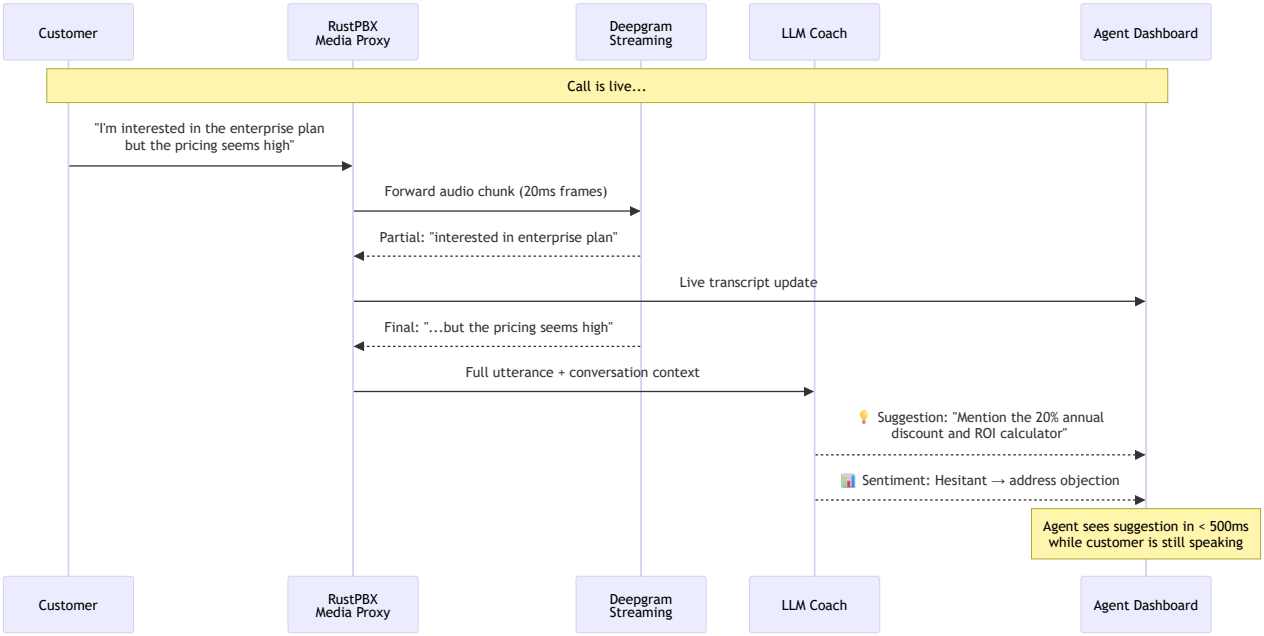
- STUN/TURN server (coturn or Telnyx TURN service)
- RustPBX WebRTC gateway (already has WebRTC support for softphone — extend to customer-facing)
- Leptos UI component for the click-to-call button widget
- Secure WebSocket signaling (wss://)
- SRTP media encryption (mandatory for WebRTC)

4.4 Own the Intelligence: Real-Time Transcription & AI Coaching

Current state: CTM provides post-call transcription only. No real-time text during live calls. No AI analysis until after the call ends.

Opportunity: Stream call audio to ASR engines in real-time during the call, feed the live transcript to an LLM for agent coaching, and surface insights while the agent can still act.





Capabilities unlocked:

- Live transcript visible to agent and supervisor during the call
- AI-suggested responses based on conversation context
- Compliance keyword detection (HIPAA, PCI, TCPA triggers)
- Automatic disposition and summary generation at call end

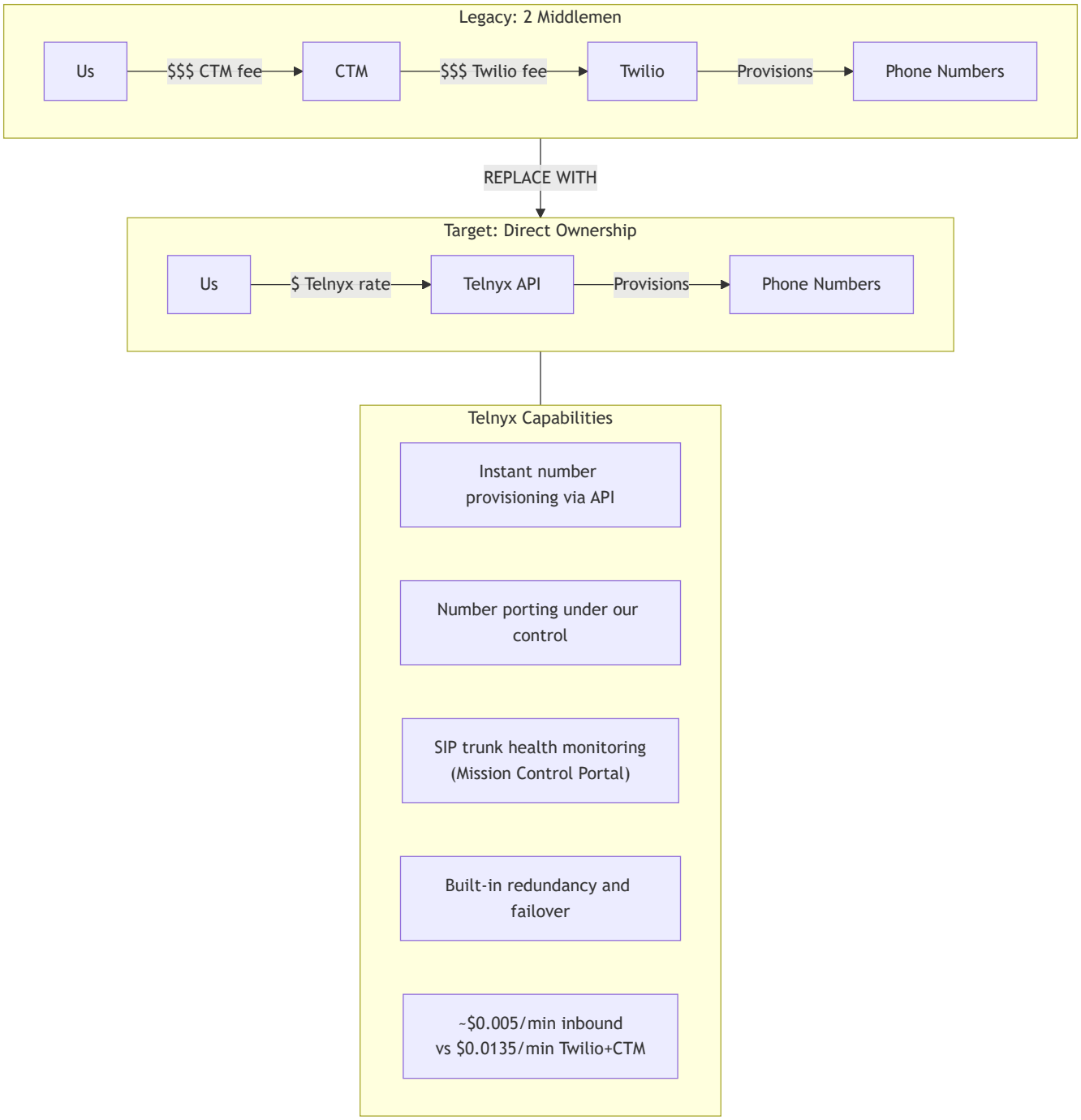
- Sentiment analysis trending during the call (escalation detection)
- Groq Whisper Turbo batch processing for final high-accuracy transcript post-call

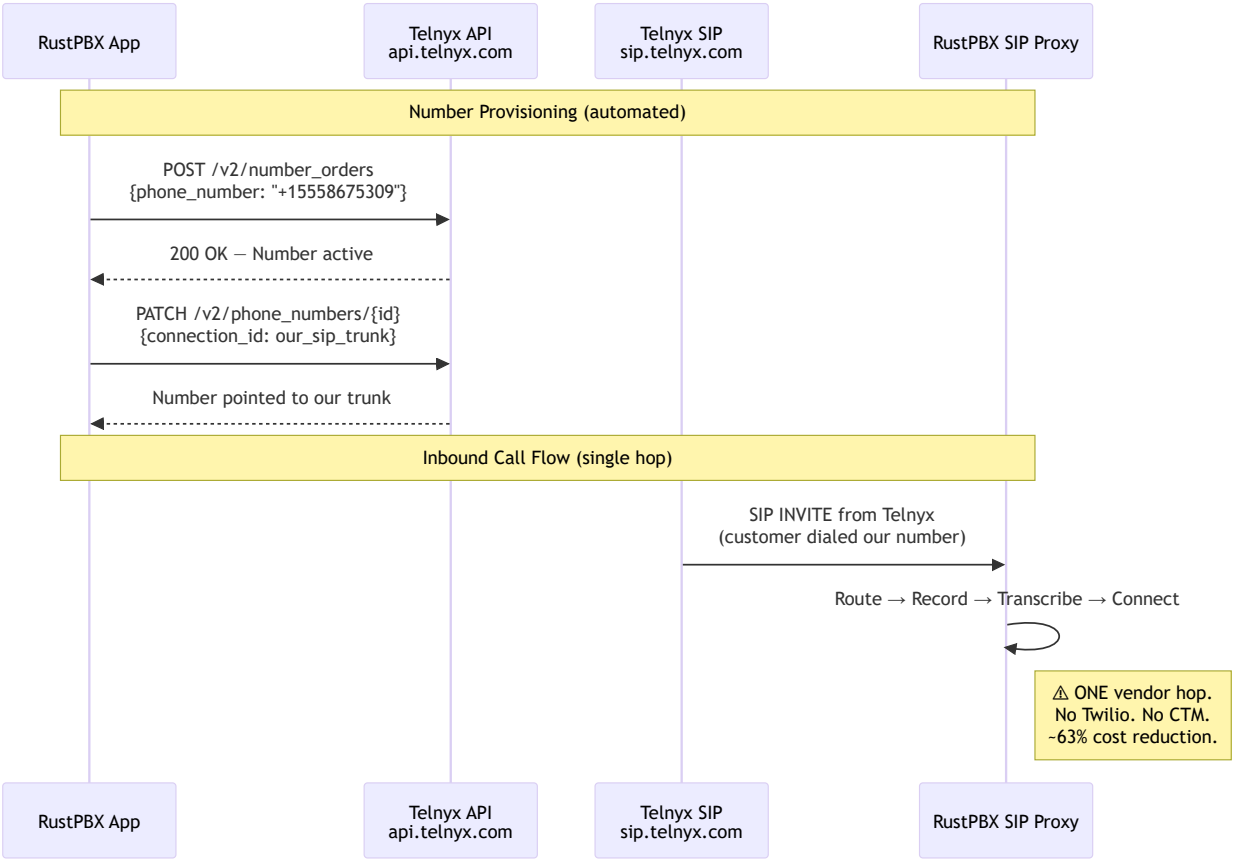
---

4.5 Own the Numbers: Direct Telnyx Integration

Current state: Tracking numbers are provisioned through CTM, which provisions them through Twilio. Two intermediaries between us and the actual phone numbers.

Opportunity: Provision and manage tracking numbers directly through Telnyx's API, cutting out both CTM and Twilio from the number management chain.


Legacy: 2 Middlemen
Us
$$$ CTM fee
CTM
$$$ Twilio fee
Twilio
Provisions
Phone Numbers
REPLACE WITH
Target: Direct Ownership
Us
$ Telnyx rate
Telnyx API
Provisions
Phone Numbers
Telnyx Capabilities
Instant number provisioning via API
Number porting under our control
SIP trunk health monitoring (Mission Control Portal)
Built-in redundancy and failover
~$0.005/min inbound vs $0.0135/min Twilio+CTM



Benefits:

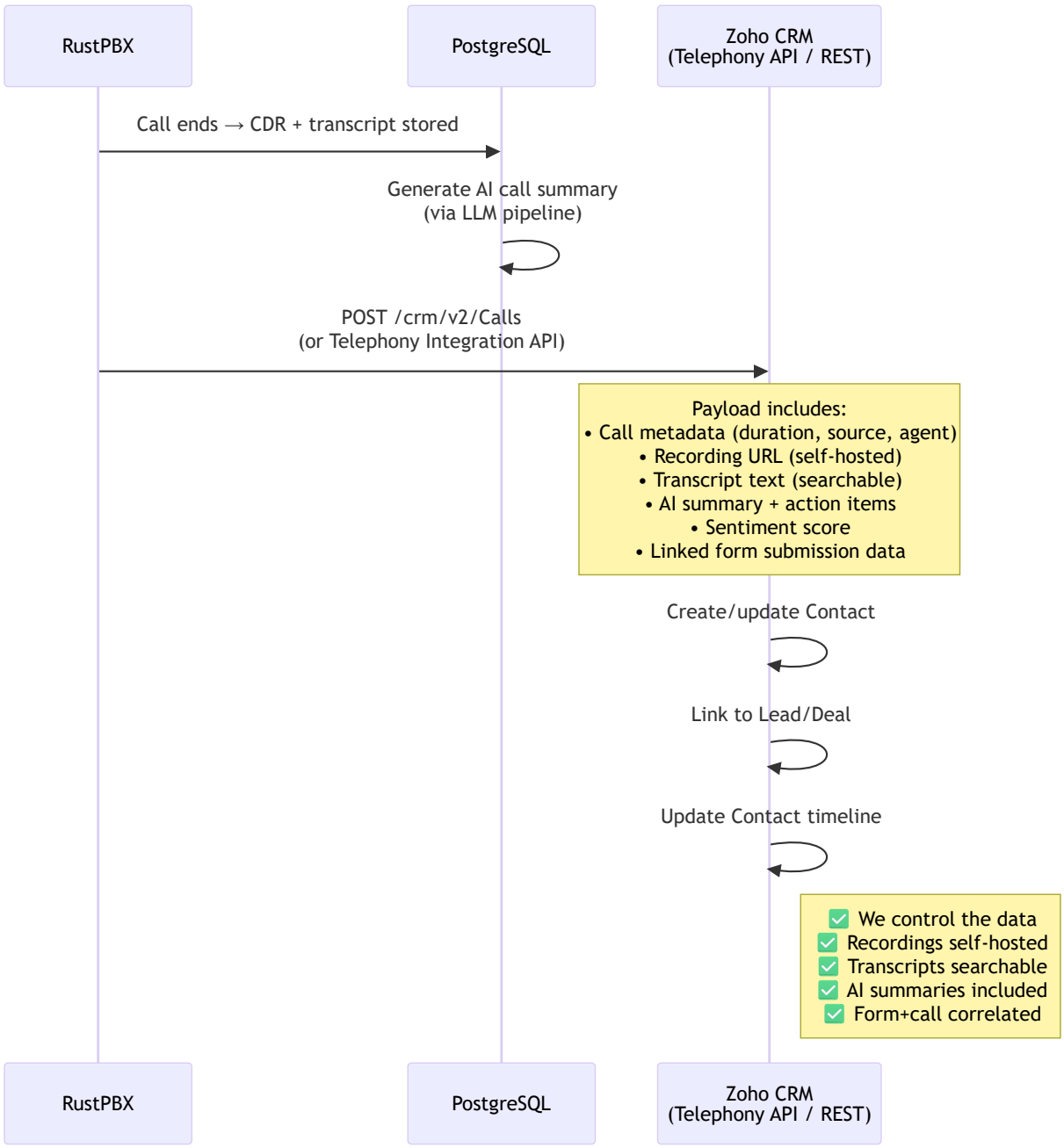
- Single SIP trunk: Telnyx → RustPBX (eliminate Twilio entirely)
- Direct number provisioning via Telnyx API (instant, programmable)
- Number porting under our control
- Per-minute cost reduction: Telnyx rates ~$0.005/min inbound vs. Twilio's combined ~$0.0135/min
- Telnyx Mission Control Portal for trunk health monitoring
- SIP trunk redundancy and failover built into Telnyx's network

---

4.6 Own the CRM Feed: Direct Zoho Integration

Current state: CTM automatically pushes call data to Zoho CRM after each call. Call metadata, recording links, tracking source, and agent info flow into Zoho contact timelines. This integration is **entirely dependent on CTM** — removing CTM severs the Zoho data pipeline.

Opportunity: Build a direct RustPBX → Zoho CRM integration that pushes richer data than CTM ever could, including transcript text, AI summaries, and correlated form submissions.



Integration approach options:

Approach	Pros	Cons
Zoho Telephony API (PhoneBridge)	Native dialer integration in Zoho UI, click-to-call	Requires Zoho marketplace registration
Zoho REST API (v2 Calls module)	Simpler, direct HTTP POST, no marketplace overhead	No native Zoho UI integration
Zoho Webhook (Notification URL)	Zoho can call us for events (new lead → screen pop)	Bidirectional, more complex

Recommended: Start with **Zoho REST API** for call data push (simplest), then evaluate **PhoneBridge Telephony API** for native click-to-call and screen pop inside Zoho.

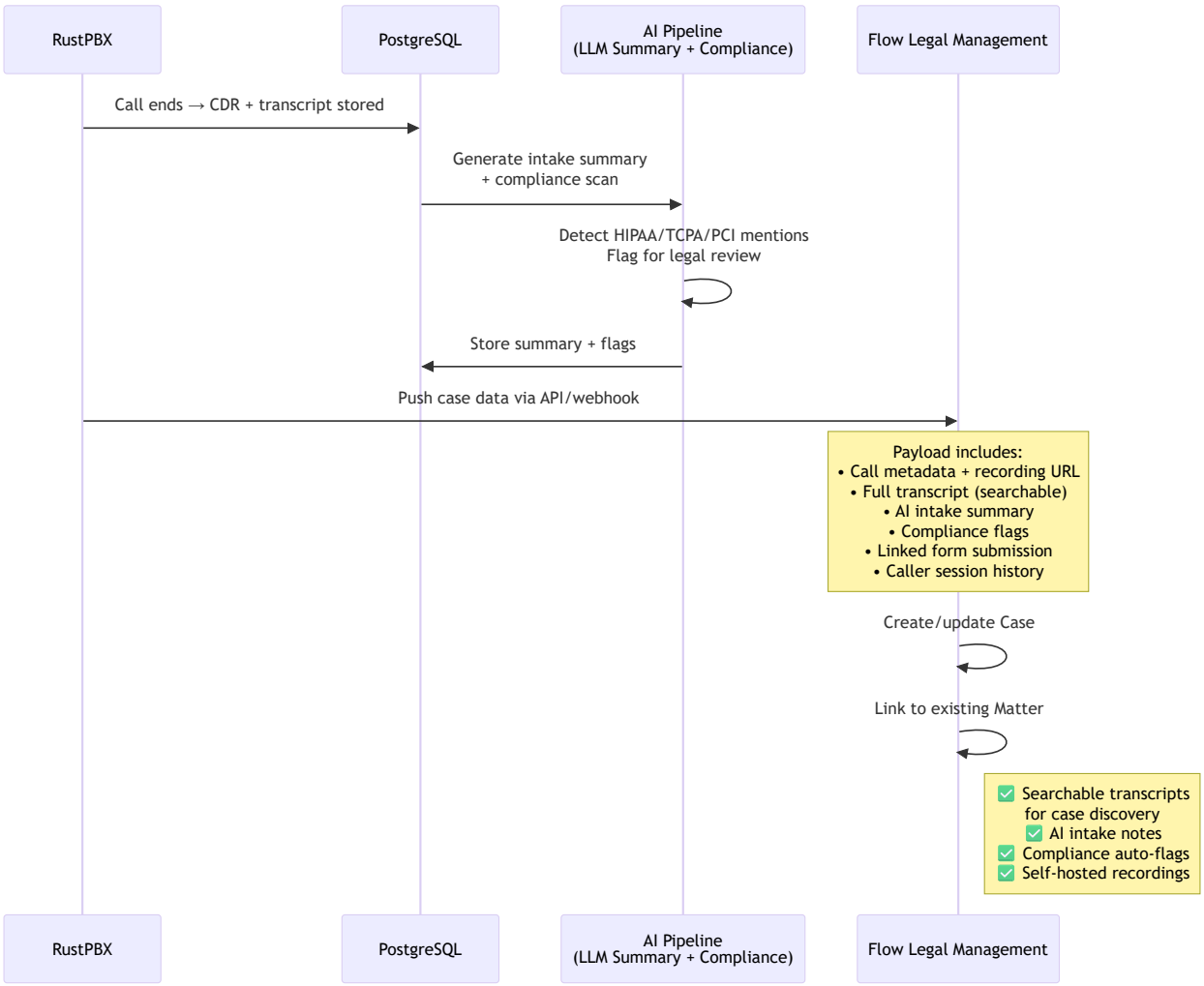
Benefits vs. CTM:

- Self-hosted recording URLs that never break (no vendor dependency)
- Searchable transcript text in CRM records
- AI-generated call summaries replace manual agent notes
- Form submissions correlated with calls via shared session_id
- Sentiment scores and compliance flags attached to each call record
- Full control over field mapping and data schema

4.7 Own the Case Pipeline: Direct Flow Legal Integration

Current state: CTM feeds call and case data into Flow Legal Management, our case management system for tracking legal matters. Like Zoho, this integration runs entirely through CTM and breaks when CTM is removed.

Opportunity: Build a direct RustPBX → Flow Legal pipeline that provides searchable transcripts, AI intake summaries, and compliance flags — data that CTM never provided.



Integration approach: This depends on Flow Legal Management's API capabilities. The integration should be built as a pluggable RustPBX addon that:

1. Listens for call-complete events from the CDR pipeline
2. Enriches call data with transcript, AI summary, and compliance flags from PostgreSQL
3. Pushes the enriched payload to Flow Legal's API endpoint
4. Handles retry logic for transient failures

Benefits vs. CTM:

- Searchable transcript text enables case discovery across all call records
- AI-generated intake summaries replace manual agent case notes
- Automatic compliance flagging (HIPAA/TCPA/PCI keyword detection)
- Self-hosted recording URLs that survive vendor changes
- Form + call correlation provides complete intake context

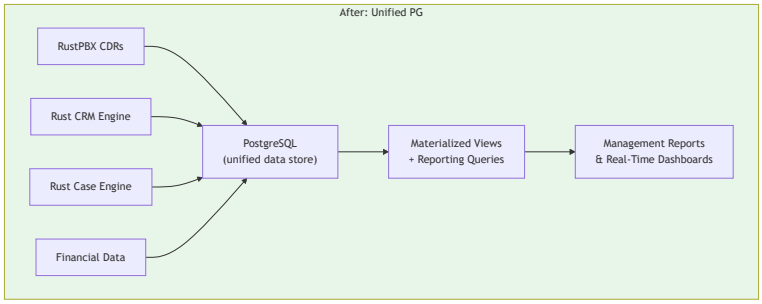
- Richer case data means less manual data entry for legal staff

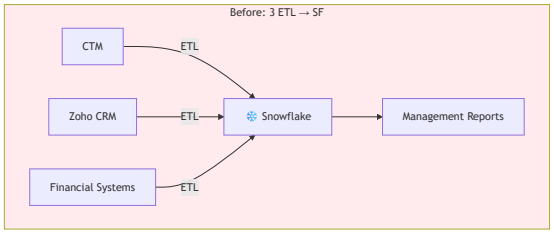
---

4.8 Own the Analytics: Replace Snowflake with PostgreSQL

Current state: Snowflake serves as the data warehouse for management reporting, aggregating ETL exports from CTM, Zoho CRM, and financial/payments systems. Reports cover call center performance, marketing attribution, and financial analytics (payments, debts, collections). Adding new cross-system metrics requires modifying ETL pipelines across multiple vendors.

Opportunity: With all call, CRM, case management, and click tracking data living in PostgreSQL (once Horizons 1–2 are complete), Snowflake becomes redundant. The same management reports can be generated directly from PostgreSQL using views, materialized views, and a lightweight reporting layer.





What this enables:

- Real-time dashboards instead of batch-lag reports (queries run against live data)
- Full-funnel analytics in one query: ad click → call → case → payment → collection
- AI-enriched analytics: sentiment trends, compliance flag rates, coaching effectiveness
- No ETL pipeline maintenance — data is already in PostgreSQL
- No Snowflake compute billing — reporting is a PostgreSQL workload
- New metrics added with a single SQL view, no multi-system pipeline changes

PostgreSQL capabilities that replace Snowflake:

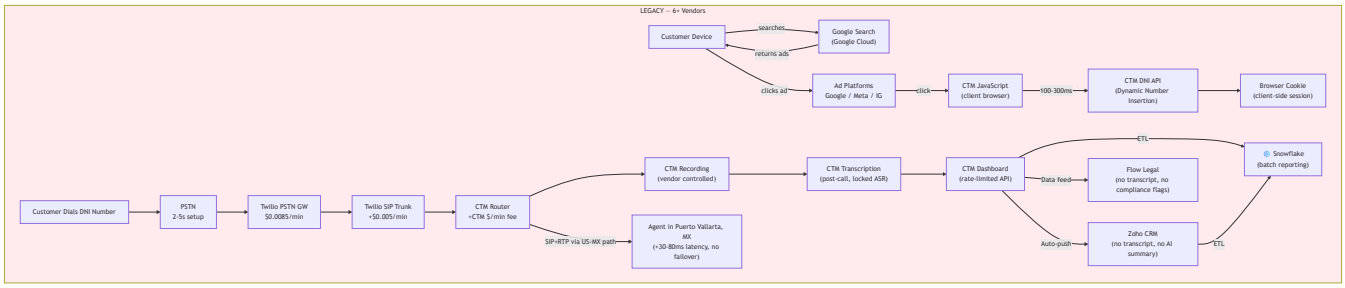
- Materialized views with `REFRESH CONCURRENTLY` for dashboard-speed aggregates
- Window functions for time-series KPIs (rolling averages, period-over-period)
- `JSONB` querying for flexible metadata analytics
- `pg_cron` for scheduled report generation
- Full-text search on transcripts for compliance auditing

Dependency: This opportunity is only viable after Zoho CRM and Flow Legal are replaced by Rust-native services in Horizon 2. Until then, Snowflake (or a transitional PostgreSQL analytics schema fed by the same ETL pipelines) continues to serve management reporting.

---

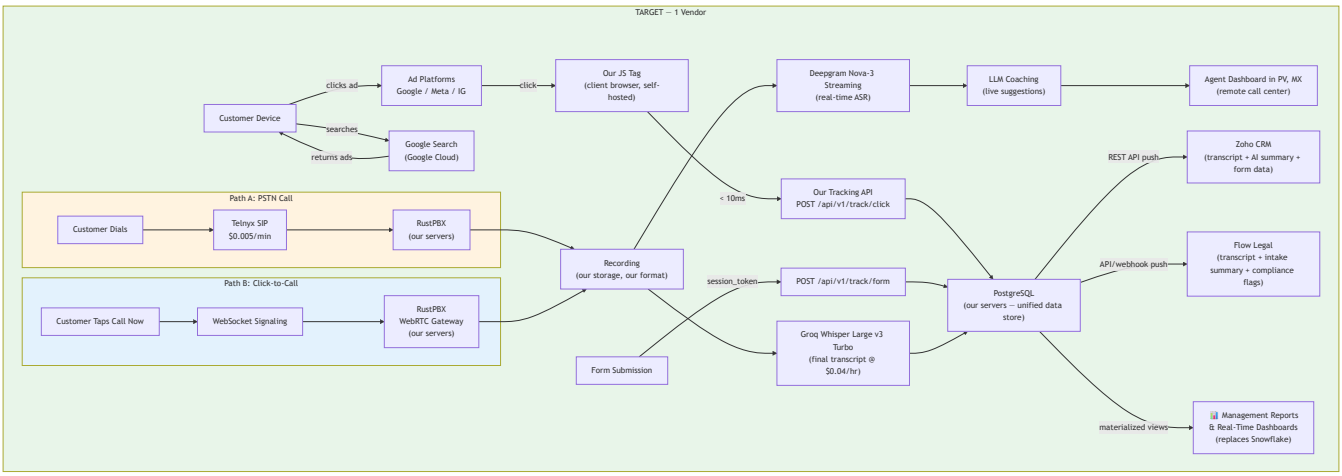
5. Target Architecture Summary

Before (Legacy)



Metric	Legacy Value
External vendors	6+ (Ad platforms, Twilio, CTM, Zoho, Flow Legal, Snowflake)
Per-call cost layers	3 (PSTN carrier, Twilio, CTM)
Data silos	7+ (each ad platform, Twilio, CTM, Zoho, Flow Legal, Snowflake, form tools)
Integration hub / SPOF	CTM (sole middleware feeding Zoho + Flow Legal + Snowflake)
Transcription control	None — CTM picks the ASR vendor
Real-time intelligence	None
Video capability	None
Call setup latency	5-10 seconds
Agent location latency	+30-80ms US-MX path, no failover
CRM transcript in records	No
Case mgmt transcript search	No
Cost per minute	~$0.02+ (Twilio + CTM combined)

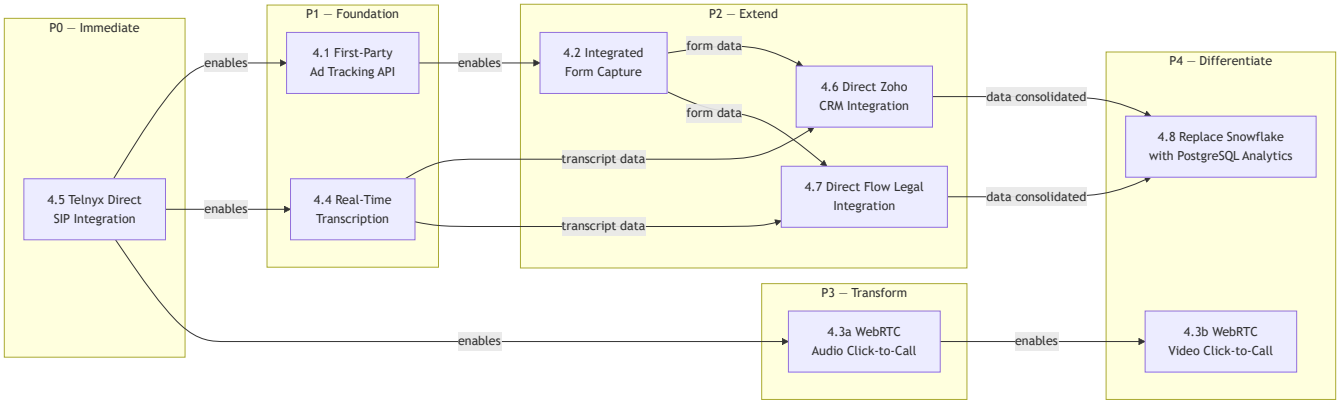
After (RustPBX Platform)



Metric	Legacy	Target	Improvement
Carrier vendors	2 (Twilio + CTM)	1 (Telnyx for PSTN only)	-50% vendor dependency
Integration hub / SPOF	CTM (feeds Zoho + Flow Legal + Snowflake)	RustPBX (direct integrations)	Eliminated single point of failure
Per-call cost layers	3	1	-67% cost overhead
Data silos	7+	0 (all in PostgreSQL, pushed to Zoho/Flow)	Unified analytics
Transcription engine	CTM-locked (no choice)	Groq Whisper Turbo + Deepgram Nova-3	Full ASR control
Real-time intelligence	None	Live transcription + AI coaching	New capability
Video capability	None	WebRTC video	New capability
Call setup latency	5-10s	<1s (WebRTC) / 3-5s (PSTN)	Up to 90% reduction
Agent failover (PV, MX)	None — ISP outage = offline	Cellular backup + regional proxy option	New capability
CRM transcript in records	No	Yes (searchable text + AI summary)	New capability
Case mgmt transcript search	No	Yes (full text + compliance flags)	New capability
Cost per minute	~$0.02+	$0.005 (PSTN) / $0.00 (WebRTC)	63-100% reduction
Management reporting	Snowflake (batch ETL, 3 vendor pipelines)	PostgreSQL views (real-time, unified)	Eliminates ETL + Snowflake billing
Full-funnel analytics	Not possible (data in 7+ silos)	Single-query: click → call → case → payment	New capability

6. Implementation Priority

Redesign opportunities ordered by impact and dependency chain:



Priority	Opportunity	Depends On	Impact
P0	4.5 — Telnyx Direct SIP Integration	RustPBX core	Eliminates Twilio cost + latency immediately
P1	4.1 — First-Party Ad Tracking API	PostgreSQL, RustPBX API	Eliminates CTM DNI dependency, owns attribution
P1	4.4 — Real-Time Transcription & Coaching	Telnyx integration, Deepgram/Groq accounts	Core differentiator vs. CTM; enables enriched CRM/case data
P2	4.2 — Integrated Form Capture	Tracking API (P1)	Unifies click + form + call data
P2	4.6 — Direct Zoho CRM Integration	Transcription (P1), Form Capture (P2)	Replaces CTM→Zoho auto-push with richer data (transcripts, AI summaries)
P2	4.7 — Direct Flow Legal Integration	Transcription (P1), Form Capture (P2)	Replaces CTM→Flow Legal feed with searchable transcripts + compliance flags
P3	4.3a — WebRTC Click-to-Call (Audio)	RustPBX WebRTC, STUN/TURN infrastructure	Eliminates PSTN for internet-connected callers
P4	4.3b — WebRTC Click-to-Call (Video)	WebRTC Audio (P3)	New capability not possible in legacy stack
P4	4.8 — Replace Snowflake with PostgreSQL Analytics	Zoho (P2), Flow Legal (P2), financial data migration	Eliminates Snowflake billing, enables real-time full-funnel reporting

> **Note on CTM removal sequencing:** CTM cannot be fully decommissioned until **both** 4.6 (Zoho) and 4.7 (Flow Legal) integrations are operational, since CTM is the sole middleware feeding both downstream systems today. Snowflake ETL pipelines from CTM must also be redirected (or Snowflake replaced per 4.8). P2 completion is the gate for CTM shutdown.

This document should be reviewed alongside ARCHITECTURE.md for the legacy system baseline, ARCHITECTURE_VISION.md for the target-state platform design, and TESTING_PLAN_OF_ACTION.md for

the testing strategy that validates each redesigned component.